

复合材料渐进损伤失效模型

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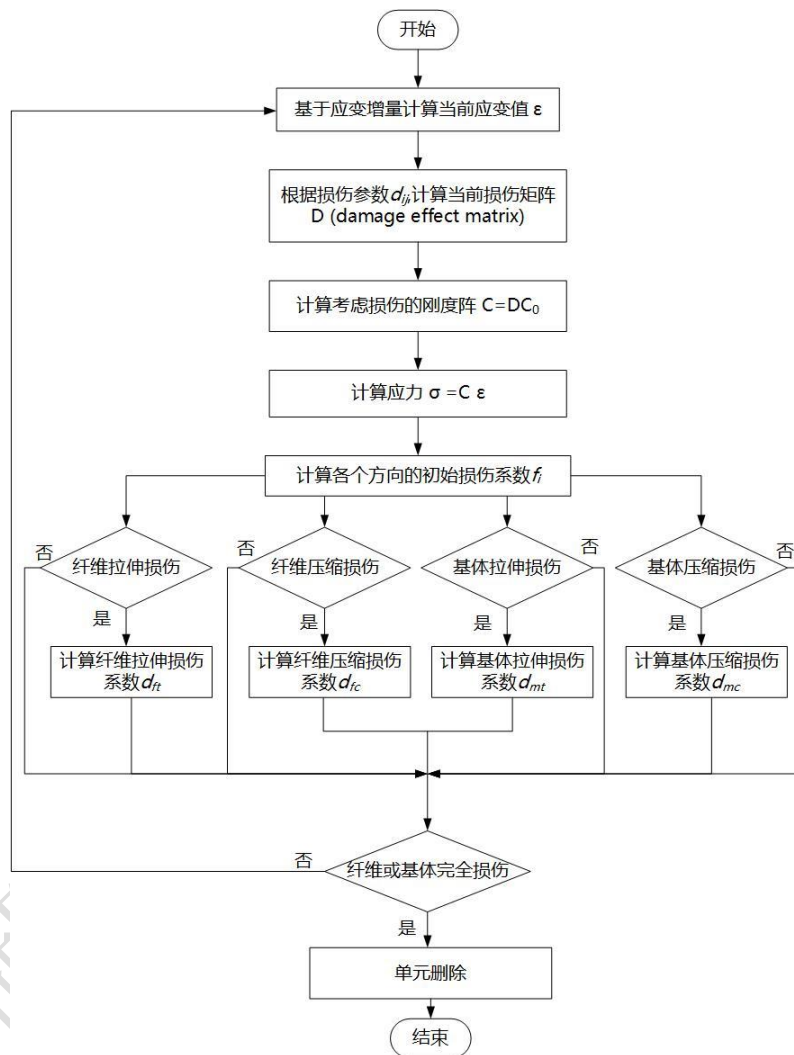


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渐进损伤失效模型课程主要内容

- (1) VUMAT 整体讲解
- (2) 在参考文献中，复合材料渐进损伤失效模型有哪些异同，有哪些关键问题。
- (3) 程序详细解读渐进损伤失效模型，包括三维 hashin 初始失效准则，渐进损伤，损伤系数如何计算，损伤矩阵（damage effect matrix）等

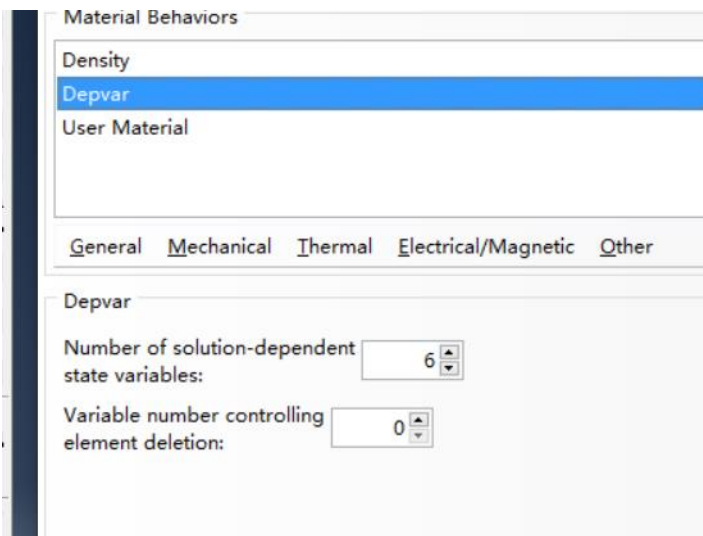


- (4) 单元测试，包括纤维拉伸和基体拉伸；如何调试子程序；对结果进行分析，包括应力，应变，初始损伤系数，损伤演化中的损伤系数，等效位移等等。
- (5) 单轴拉伸模型的建立与结果分析，与 abaqus 自带的二维 hashin 和渐进损伤对比。
- (6) 模型的改进与结果分析

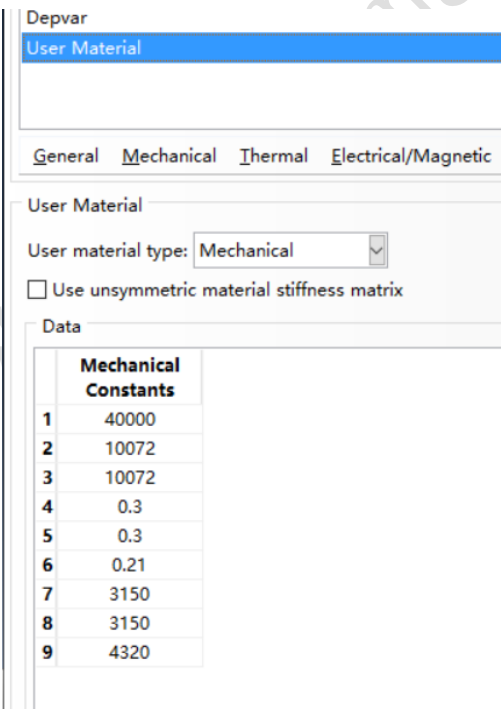


1 VUMAT

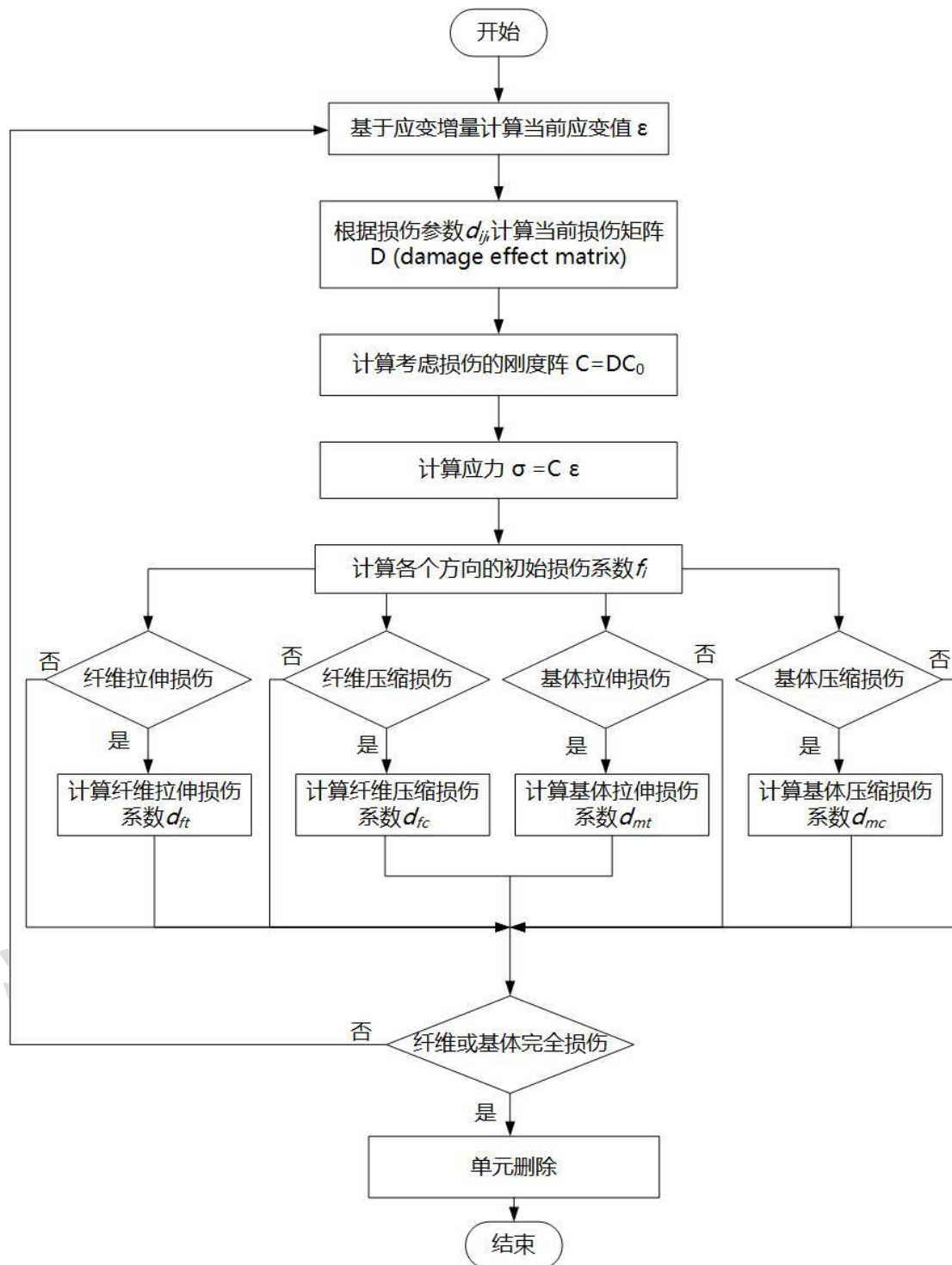
user-defined material subroutine 没有的材料模型
基于当前应变增量和上一步的应变，通过 VUMAT 得出当前的应力值。
输入：当前应变增量，上一步的应变
输出：可以自己设定输出值，stateNew，如下为设置 6 个输出值。



材料定义：以下值对应 VUMAT 中的 props(i)，需要输入的材料参数。



2 参考文献



2.1 基于应力描述的三维 hashin 损伤初始准则

2.1.1 Ref1

Hashin's 3D failure criteria are sufficient to simulate woven glass-fibre/polypropylene composite layer. The failure functions may be expressed as follows^[1]:

$$\text{Fibre tension } (\sigma_{11} \geq 0): F_f^t = \left(\frac{\sigma_{11}}{X_{1t}}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2, d_{ft} = 1$$

$$\text{Fibre compression } (\sigma_{11} < 0): F_f^c = \frac{|\sigma_{11}|}{X_{1c}}, d_{fc} = 1$$

$$\text{Matrix tension } (\sigma_{22} + \sigma_{33} \geq 0): F_m^t = \frac{(\sigma_{22} + \sigma_{33})^2}{X_{2t}^2} + \frac{\sigma_{23}^2 - \sigma_{22}\sigma_{33}}{S_{23}^2} + \frac{\sigma_{12}^2 + \sigma_{13}^2}{S_{12}^2}, d_{mt} = 1$$

$$\text{Matrix compression } (\sigma_{22} + \sigma_{33} < 0):$$

$$F_m^c = \left[\left(\frac{X_{2c}}{2S_{23}}\right)^2 - 1 \right] \frac{(\sigma_{22} + \sigma_{33})^2}{X_{2c}^2} + \frac{(\sigma_{22} + \sigma_{33})^2}{4S_{23}^2} + \frac{(\sigma_{23}^2 - \sigma_{22}\sigma_{33})}{S_{23}^2} + \frac{\sigma_{12}^2 + \sigma_{13}^2}{S_{12}^2}, d_{mc} = 1 \quad (7)$$

where $X_{1t}, X_{1c}, X_{2t}, X_{2c}, S_{12}, S_{13}$ and S_{23} are the various strength components [18] and d_{ft}, d_{fc}, d_{mt} and d_{mc} are the damage variables associated with the four failure modes.

2.1.2 Ref2

The Hashin failure criteria are improved for the woven composite laminates considering the delamination, warp and weft fiber failures. The modified Hashin criteria are presented as below^[2]:

$$\text{Warp axial tensile-shear: } \left(\frac{\sigma_{11}}{X_T}\right)^+ + \left(\frac{\sigma_{12}}{S_{12}}\right)^+ + \left(\frac{\sigma_{13}}{S_{13}}\right)^+ \geq 1, \sigma_{11} \geq 0 \quad (1)$$

$$\text{Warp compressive warping: } \left(\frac{\sigma_{11}}{X_C}\right)^- \geq 1, \sigma_{11} < 0 \quad (2)$$

$$\text{Weft axial tensile-shear: } \left(\frac{\sigma_{22}}{Y_T}\right)^+ + \left(\frac{\sigma_{12}}{S_{12}}\right)^+ + \left(\frac{\sigma_{23}}{S_{23}}\right)^+ \geq 1, \sigma_{22} \geq 0 \quad (3)$$

$$\text{Weft compressive warping: } \left(\frac{\sigma_{22}}{Y_C}\right)^- \geq 1, \sigma_{22} < 0 \quad (4)$$

$$\text{Delamination: } \left(\frac{\sigma_{33}}{Z_T}\right)^+ + \left(\frac{\sigma_{13}}{S_{13}}\right)^+ + \left(\frac{\sigma_{23}}{S_{23}}\right)^+ \geq 1, \sigma_{33} \geq 0 \quad (5)$$

$$\left(\frac{\sigma_{13}}{S_{13}}\right)^- + \left(\frac{\sigma_{23}}{S_{23}}\right)^- \geq 1, \sigma_{33} < 0 \quad (6)$$

Where: X_T, Y_T are tensile strength of the warp and weft fiber; X_C, Y_C are compressive strength of the warp and weft fiber; S_{12}, S_{13}, S_{23} are shear strength. The corresponding stress components could be set zero while elements fail.

2.1.3 Ref3

Hashin criterion is one of the most famous criteria for predicting failure initiation



in one layer of composite laminates, which is established in terms of expressions using material strengths. The four different failure modes in Hashin criterion are given as follows^[3]:

Fiber tensile failure ($\sigma_{11} \geq 0$)

$$F_{ft} = \left(\frac{\sigma_1}{X_T}\right)^2 + \alpha \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \alpha \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1 \quad (1)$$

Fiber compressive failure ($\sigma_{11} < 0$)

$$F_{fc} = \left(\frac{\sigma_1}{X_C}\right)^2 \geq 1 \quad (2)$$

Matrix tensile failure ($\sigma_{22} + \sigma_{33} \geq 0$)

$$F_{mt} = \left(\frac{\sigma_2 + \sigma_3}{Y_T}\right)^2 + \left(\frac{1}{S_{23}^2}\right)(\sigma_{23}^2 - \sigma_{22}\sigma_{33}) + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1 \quad (3)$$

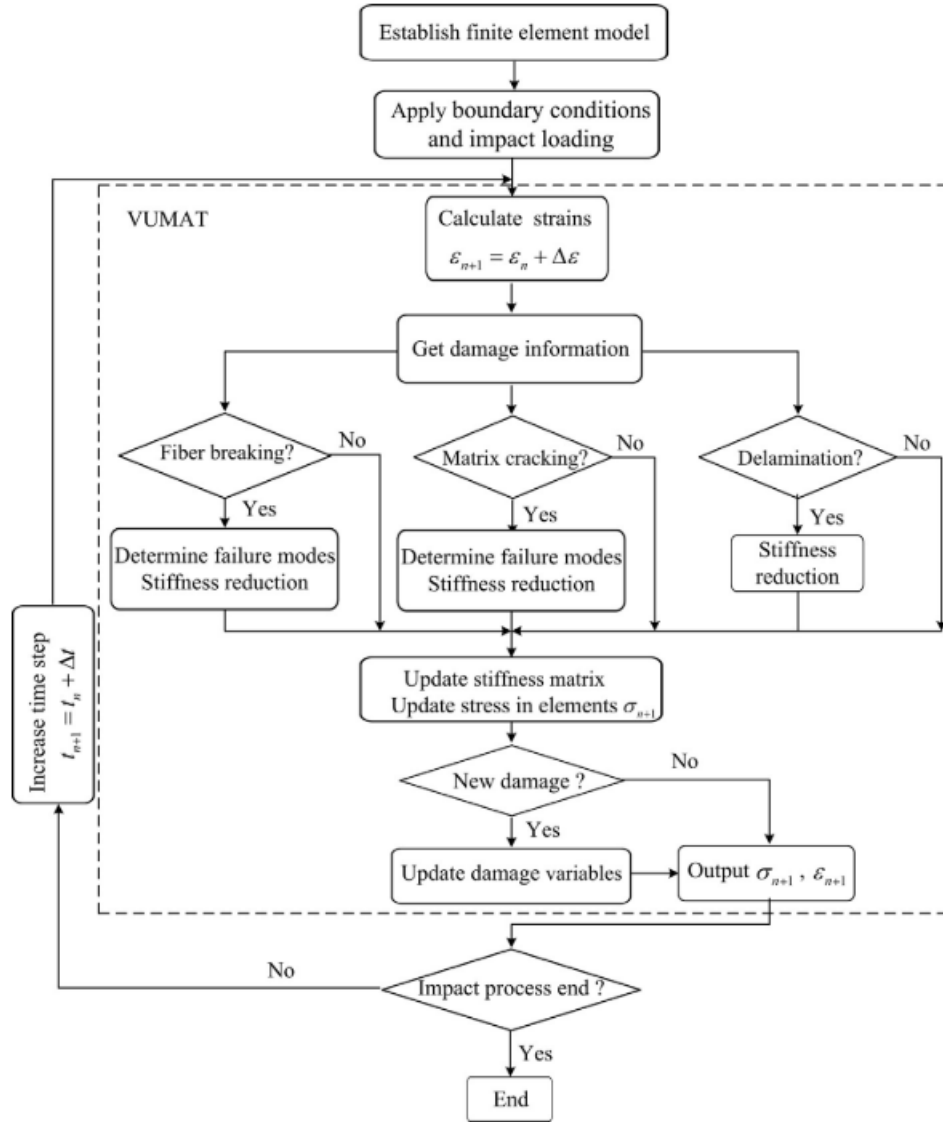
Matrix compressive failure ($\sigma_{22} + \sigma_{33} < 0$)

$$F_{mc} = \left(\frac{\sigma_2 + \sigma_3}{2S_{23}}\right)^2 + \left(\frac{\sigma_2 + \sigma_3}{Y_C}\right) \left[\left(\frac{Y_C}{2S_{23}}\right)^2 - 1 \right] + \frac{1}{S_{23}^2}(\sigma_{23}^2 - \sigma_2\sigma_3) + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1 \quad (4)$$

In the above equations, X_T and X_C are the tensile and compressive strengths in fiber direction; Y_T and Y_C are the tensile and compressive strengths in transverse direction; S_{12} , S_{13} and S_{23} are the shear strengths respectively; α is the shear failure coefficient applied to determine the contribution of shear stresses on the fiber tensile failure.

Based on the corresponding constitutive models, the stresses and damage states on the layers and interfaces can be obtained. Once the failure initiation criterion is reached, the stiffness reduction is carried out by updating the damage variables, and then the stresses at the integration points of elements are updated by the reduced stiffness matrix. At last, the updated state variables are returned to ABAQUS to push next increment analysis until the end of the impact process. The flow chart of the numerical modeling process is presented in details in Fig. 2^[3].





The material properties of unidirectional composite and interface cohesive elements used in the simulation are cited from Refs. [21,29] and summarized in Tables 1 and 2^[3]

Table 1
Material properties of unidirectional composite.

Density (kg/m ³)	1600
Stiffness properties	$E_{11} = 153 \text{ GPa}$; $E_{22} = E_{33} = 10.3 \text{ GPa}$; $\nu_{12} = \nu_{13} = 0.3$; $\nu_{23} = 0.4$ $G_{12} = G_{13} = 6 \text{ GPa}$; $G_{23} = 3.7 \text{ GPa}$
Strength properties (MPa)	$X_T = 2537$; $X_C = 1580$; $Y_T = 82$; $Y_C = 236$; $S_{12} = S_{13} = 90$; $S_{23} = 40$
Fracture energy (N/mm)	$G_{ft} = G_{fc} = 10$; $G_{mt} = G_{mc} = 1$



Table 2
Material properties of interface cohesive elements.

K (N/mm ³)	N (MPa)	$S = T$ (MPa)	G_{IC} (N/mm)	$G_{IIC} = G_{IIIC}$ (N/mm)	Density (kg/m ³)
2×10^5	62.3	92.3	0.28	0.79	1200

2.1.4 Ref4

The famous Hashin failure criteria described in reference of Zhang and Zhang (2015) are chosen to track intra-laminar matrix and fiber damages and summarized as follow^[4]:

Tensile matrix cracking

$$\left(\frac{\sigma_{22}}{Y_T}\right)^2 + \left(\frac{\tau_{12}}{S_{12}}\right)^2 + \left(\frac{\tau_{23}}{S_{23}}\right)^2 \geq 1 \quad (1)$$

Compressive matrix cracking

$$\left(\frac{\sigma_{22}}{Y_C}\right)^2 + \left(\frac{\tau_{12}}{S_{12}}\right)^2 + \left(\frac{\tau_{23}}{S_{23}}\right)^2 \geq 1 \quad (2)$$



Tensile fiber failure

$$\left(\frac{\sigma_{11}}{X_T}\right)^2 + \left(\frac{\tau_{12}}{S_{12}}\right)^2 + \left(\frac{\tau_{23}}{S_{23}}\right)^2 \geq 1 \quad (3)$$

Compressive fiber failure

$$\left(\frac{\sigma_{11}}{X_T}\right)^2 + \left(\frac{\tau_{12}}{S_{12}}\right)^2 + \left(\frac{\tau_{23}}{S_{23}}\right)^2 \geq 1 \quad (4)$$

Fiber-matrix shear-out

$$\left(\frac{\sigma_{11}}{X_C}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{23}}{S_{23}}\right)^2 \geq 1 \quad (5)$$

To satisfy the intra-laminar failure criteria for damage evolution, material degradation rule (Zhang and Zhang 2015) is applied to degrade the structural stiffness of the material properties in order to compute the actual impact threshold. The degradation material properties are listed in Table 1.

Where σ_{11} and σ_{22} are the in-plane stresses in the fiber and transverse directions, τ_{12} and τ_{23} represent the shear stresses, X_T and X_C refer to the fiber tension and compression strengths, Y_T and Y_C are the matrix tension and compression strengths, S_{XY} is the fiber-matrix shear-out, S_{12} and S_{23} denote the shear strengths, and 'd' refers to the degraded material properties. The E and G denote allowable elastic and shear modulus, respectively.

2.1.5 Ref5

Failure in the 3D woven composite can be modelled using the quadratic stress-based criteria described in Gama and Gillespie (2011). Essentially, these are the generalizations of the failure criteria proposed by Hashin (1980) for a unidirectional composite. There are four failure conditions which are described below^[5].

Tensile-shear failure in the warp fibre direction:

$$f_{1t} = \left(\frac{\sigma_{11}}{S_{1t}}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{31}}{S_{31}}\right)^2 - 1 = 0, \quad \sigma_{11} > 0. \quad (5)$$

Tensile-shear failure in the weft fibre direction:



$$f_{2t} = \left(\frac{\sigma_{22}}{S_{2t}} \right)^2 + \left(\frac{\sigma_{12}}{S_{12}} \right)^2 + \left(\frac{\sigma_{23}}{S_{23}} \right)^2 - 1 = 0, \quad \sigma_{22} > 0. \quad (6)$$

Compressive failure in the warp fibre direction:

$$f_{1c} = \left(\frac{\sigma_{11}}{S_{1c}} \right)^2 - 1 = 0, \quad \sigma_{11} < 0. \quad (7)$$

Compressive failure in the weft fibre direction:

$$f_{2c} = \left(\frac{\sigma_{22}}{S_{2c}} \right)^2 - 1 = 0, \quad \sigma_{22} < 0. \quad (8)$$

Instant failure implies that once one of Eqs. (5)–(8) is satisfied, the material is assumed to fail and the element is removed from the mesh.

The VUMAT subroutine that describes the composite constitutive response and failure model was incorporated into the main Abaqus program. The subroutine is called at each time increment to perform numerical integrations for the composite. The values for the time and strain increments are calculated automatically through the Abaqus/Explicit main program. Strain increments are used to compute stress increment values and subsequently update the stress values from the previous step. A flowchart describing the implementation of the instant failure model in the VUMAT subroutine is presented in Fig. 1



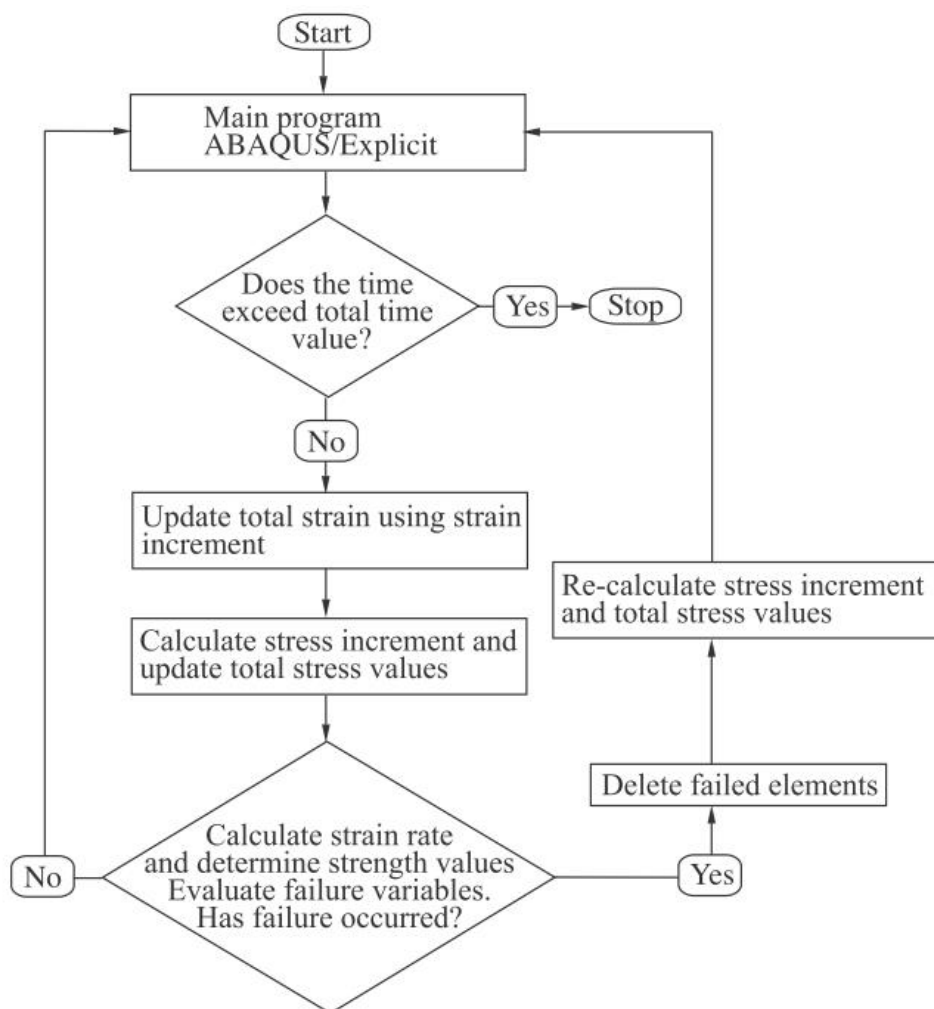


fig. 1. Flowchart of the user-defined subroutine for calculating the blast response of the composite material.

2.2 损伤演化

重点问题:

damage effect tensor D 表达方式 不同文献表达方式不同

damage variable 表达方式

df , dft , dfc

dm , dmt , dmc

应变, 位移

初始损伤准则只有 hashin 的文献^{[3][6][7]}

含有 hashin 的混合: 文献^{[8][9]}

指数损伤演化: ^[10]

PUCK 初始损伤 ^[11]



损伤演化分为两类^[9]，同时发现损伤演化与初始损伤准则存在一定的联系，如初始损伤准则采用 Puck，那么损伤演化也会和 PUCK 有关（fracture plane）。

Fracture energy-based damage evolution methods with linear degradation trends are often classified into two broad types, **equivalent strain** or **equivalent displacement**. In the equations below, i represents the damage mode and j denotes the longitudinal or transverse direction of the laminate.

2.1.3.1. Equivalent strain method.

$$d_i = \frac{\varepsilon_{eq,i}^f (\varepsilon_{eq,i} - \varepsilon_{eq,i}^0)}{\varepsilon_{eq,i} (\varepsilon_{eq,i}^f - \varepsilon_{eq,i}^0)} \quad (11)$$

with $\varepsilon_{eq,i}^0 = X_i/E_{0,j}$, $\varepsilon_{eq,i}^f = 2G_i/(X_i l_c)$, where $\varepsilon_{eq,i}^0$ and $\varepsilon_{eq,i}^f$ represent the initial and final failure strain, respectively; X_i is the fiber or matrix strength; $E_{0,j}$ is the initial modulus of the composite lamina in the corresponding direction; G_i is the critical fracture energy for the specific damage mode; and l_c is the characteristic length of the element, which is introduced to eliminate the mesh sensitivity. In the present study, l_c is considered to be equal to the cube root of the element volume.

2.1.3.2. Equivalent displacement method. The equivalent displacement method can be described by the following equation.

$$d_i = \frac{\delta_{eq,i}^f (\delta_{eq,i} - \delta_{eq,i}^0)}{\delta_{eq,i} (\delta_{eq,i}^f - \delta_{eq,i}^0)} \quad (12)$$

in which $\delta_{eq,i}^0 = \delta_{eq,i}/\sqrt{r_i}$ and $\delta_{eq,i}^f = 2G_i/\sigma_{eq,i}^0$, where $\delta_{eq,i}^0$ and $\delta_{eq,i}^f$ denote the initial and fully damaged equivalent displacements, respectively; r_i is the value of the corresponding failure criterion; G_i is the critical fracture energy for the specific damage mode, the same as in Eq. (11); and $\sigma_{eq,i}^0$ is the equivalent stress at the instant of damage initiation. Detailed expressions for calculating the equivalent displacement and equivalent stress for each failure mode can be found in Ref. [33].

2.2.1 REF^[11] PUCK+ equivalent strain

components of the effective stress tensor $\tilde{\sigma}$ are related to the actual Cauchy stress tensor σ , expressed via the damage effect tensor D , undamaged material elasticity tensor C and strain tensor ε ,

$$\tilde{\sigma} = D\sigma = DC\varepsilon. \quad (1)$$



Damage effect tensor D :

The damage effect tensor D is constructed as follows:

$$D = \text{diag} \left[\frac{1}{1-d_{11}}, \frac{1}{1-d_{22}}, \frac{1}{1-d_{33}}, \frac{1}{1-d_{23}}, \frac{1}{1-d_{13}}, \frac{1}{1-d_{12}} \right] \quad (2)$$

and

$$\begin{cases} d_{11} = 1 - (1 - d_{11}^t)(1 - d_{11}^c) \\ d_{22} = 1 - (1 - d_{22}^t)(1 - d_{22}^c) \\ d_{33} = 1 - (1 - d_{33}^t)(1 - d_{33}^c) \end{cases} \quad (3)$$

where damage variables defined by d_i^k ($i = 1, 2, 3$ and $k = t, c$) modify the material stiffness to represent the intralaminar progressive damage and account for damage irreversibility. Actually, these constitutive laws for damaged material originated from the principle of strain equivalence [14]; when $d_i^k = 0$, the material is intact, and when $d_i^k = 1$, the material completely fails. Each damage variable is able to represent a type of intralaminar damage mode, and the following damage modes are considered:

- Longitudinal damage under tension d_{11}^t , i.e., fiber tensile failure.
- Longitudinal damage under compression d_{11}^c , i.e., fiber compressive failure.
- Transverse damage under tension d_{22}^t/d_{33}^t , i.e., matrix tensile failure.
- Transverse damage under compression d_{22}^c/d_{33}^c , i.e., matrix compressive failure.
- Shear damage d_{ij} , for $i, j = 1, 2, 3$ and $i \neq j$, i.e., matrix cracking due to shear loading.

To illustrate the irreversibility of the damage, the damage variable is defined by the following function of analysis time t and incremental time Δt ,

$$d_{ii}(t + \Delta t) = \max\{0, \min\{1, d_{ii}(t + \Delta t)\}\}, i = 1, 2, 3. \quad (4)$$

使得 damage variable 在[0,1]范围内变化, 同时还需要保证损伤不可逆^[12],

To account for irreversibility, the damage variable as a function of analysis time, t , defined as

$$\begin{cases} d_{ii}(t + \Delta t) = \max\{0, \min\{1, d_{ii}(t + \Delta t)\}\} \\ d_{ii}(t + \Delta t) \geq d_{ii}(t) \end{cases}, i = 1, 2, 3. \quad (4)$$

Damage variable:

Once the failure criterion has been met, a bilinear evolution law was used to model



the gradual reduction in material properties

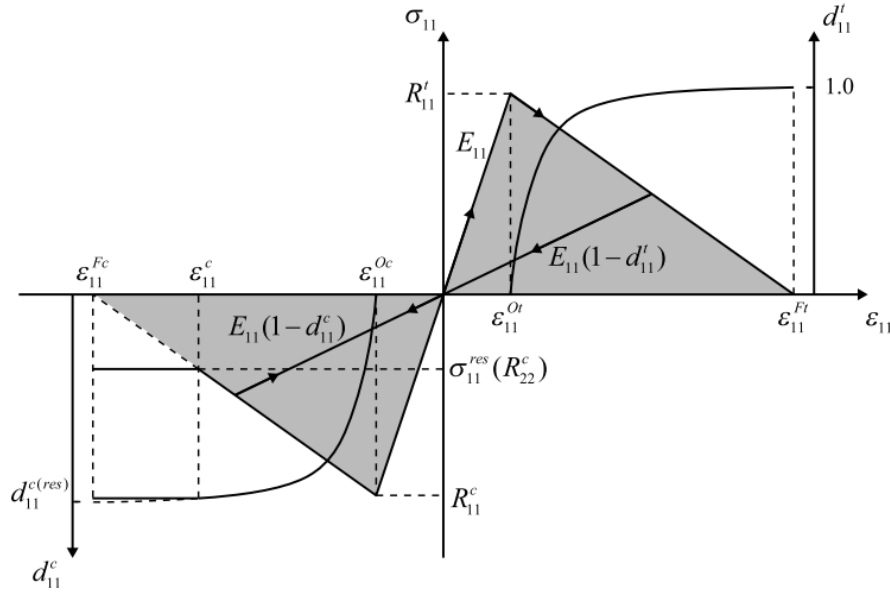


Fig. 1. Constitutive bilinear evolution law for longitudinal damage (shaded area is fracture energy $G'_{1c(c)}$).

纤维拉伸和纤维压缩

Once the failure criterion has been met, a bilinear evolution law (Fig. 1) was used to model the gradual reduction in material properties for tensile and compressive failure in the longitudinal direction, and the damage variable is given by the following equation:

$$d'_{11(c)} = \frac{\epsilon_{11}^{Ft(c)}}{\epsilon_{11}^{Ft(c)} - \epsilon_{11}^{Or(c)}} \left(1 - \frac{\epsilon_{11}^{Or(c)}}{\epsilon_{11}^{Ft(c)}} \right) \quad (7)$$

where the failure initiation strain $\epsilon_{11}^{Or(c)}$ (t and c for tension and compression, respectively) was determined by the ultimate tensile and compressive strengths ($R'_{11(c)}$) in the longitudinal direction, i.e., $\epsilon_{11}^{Or(c)} = R'_{11(c)} / C_{11}$, and $\epsilon_{11}^{Ft(c)}$ correspond to the maximum strain at which the damage variables reach unity. For the sake of achieving mesh objectivity, the characteristic length l^* is introduced to solve the maximum strain at failure by regularizing the fiber fracture energy $G'_{1c(c)}$ [29]. The expression is given by:

$$\epsilon_{11}^{Ft(c)} = \frac{2G'_{1c(c)}}{R'_{11(c)} l^*} \quad (8)$$

Fracture energy^[11]/ Fracture energy density^[3]/fracture toughness^{[8][12]}的单位是 N/mm，纵坐标是应力，横坐标是位移，君莫课件如下：



如何计算特征长度? characteristic length^[12]

An accurate measure of the characteristic length is the ratio of the elemental volume V and fracture plane area A ,

$$l_{fib} = \frac{V}{A} \quad (10)$$

where A is calculated using an approach proposed in [21] and described in Section 2.1.2.

根据 abaqus 的帮助文档: charLength(nblock)表示单元的特征长度, 对于一阶单元, 四边形单元为面积的开方, 六面体单元为体积的开三次方^[9]。

Abaqus user subroutines reference guide>1.2 abaqus/Explicit subroutines>1.2.22 VUMAT

charLength(nblock)

Characteristic element length, which is either the default value based on the geometric mean or the user-defined characteristic element length defined in user subroutine VUCHARLENGTH. **The default value is a typical length of a line across an element for a first-order element;** it is half of the same typical length for a second-order element. For beams, pipes, and trusses, the default value is a characteristic length along the element axis. For membranes and shells it is a characteristic length in the reference surface. For axisymmetric elements it is a characteristic length in the r-z plane only. For cohesive elements it is equal to the constitutive thickness.

基体拉伸和压缩如下, 过于较为复杂



When matrix damage initiation occurs, the matrix damage evolution parameter $d_{mat}^{t(c)}$ on the fracture plane can be effectively predicted by the corresponding equivalent strain:

$$d_{mat}^{t(c)} = \frac{\varepsilon_{eq}^{Fr(c)}}{\varepsilon_{eq}^{Fr(c)} - \varepsilon_{eq}^{Or(c)}} \left(1 - \frac{\varepsilon_{eq}^{Or(c)}}{\varepsilon_{eq}^{t(c)}} \right) \quad (13)$$

where $\varepsilon_{eq}^{t(c)}$, $\varepsilon_{eq}^{Or(c)}$ and $\varepsilon_{eq}^{Fr(c)}$ are the equivalent strains at the current moment, damage initiation, and complete damage, respectively. $\varepsilon_{eq}^{t(c)}$ and $\varepsilon_{eq}^{Or(c)}$ can be represented by a combination of the strain components ε_n , ε_{nt} and ε_{n1} on the fracture plane, which are obtained by rotating the strain ε_{ij} ($i, j = 1, 2, 3$) in the material system to the fracture plane, as follows:

$$\begin{cases} \varepsilon_{eq}^{t(c)} = \sqrt{\langle \varepsilon_n \rangle^2 + (\varepsilon_{nt})^2 + (\varepsilon_{n1})^2} \\ \varepsilon_{eq}^{Or(c)} = \sqrt{\langle \varepsilon_n^0 \rangle^2 + (\varepsilon_{nt}^0)^2 + (\varepsilon_{n1}^0)^2} \end{cases} \quad (14)$$

对于基体方向的损伤，22 和 33 如何处理？损伤演化中的 damage variable 与刚度退化的联系

Transverse damage under tension d_{22}^t/d_{33}^t , i.e., matrix tensile failure

Transverse damage under compression d_{22}^c/d_{33}^c , i.e., matrix compressive failure

当发生基体拉伸损伤时， d_{22}^t d_{33}^t 均发生改变，从而造成刚度 C_{22} 和 C_{33} 的改变。

$$D = \text{diag} \left[\frac{1}{1-d_{11}}, \frac{1}{1-d_{22}}, \frac{1}{1-d_{33}}, \frac{1}{1-d_{23}}, \frac{1}{1-d_{13}}, \frac{1}{1-d_{12}} \right] \quad (2)$$

and

$$\begin{cases} d_{11} = 1 - (1-d_{11}^t)(1-d_{11}^c) \\ d_{22} = 1 - (1-d_{22}^t)(1-d_{22}^c) \\ d_{33} = 1 - (1-d_{33}^t)(1-d_{33}^c) \end{cases} \quad (3)$$

$$\tilde{\sigma} = D\sigma = DC\varepsilon. \quad (1)$$

2.2.2 Ref^[3] equivalent displacement



2.1.2. Damage evolution law

Once the damage initiation criterion is reached, the damage development requires a damage evolution law. Herein, a gradual degradation scheme coupling with Murakami damage model [37] proposed by Lapczyk et al. [38] and Fang et al. [39] is applied to characterize the intra-laminar damage evolution. The damage variable for each failure mode is expressed as:

$$d_I = \frac{\delta_{I,eq}^f (\delta_{I,eq} - \delta_{I,eq}^0)}{\delta_{I,eq} (\delta_{I,eq}^f - \delta_{I,eq}^0)} \quad (d_I \in [0, 1], \quad I = ft, fc, mt, mc) \quad (5)$$

where $\delta_{I,eq}^0$ is the equivalent displacement for damage initiation; $\delta_{I,eq}^f$ is the equivalent displacement at final failure; and they can be determined by

$$\delta_{I,eq}^0 = \delta_{I,eq} / \sqrt{F_I} \quad (6)$$

$$\delta_{I,eq}^f = 2G_I / (\sigma_{I,eq} / \sqrt{F_I}) \quad (7)$$

where F_I is the value of damage initiation criterion; G_I is the fracture energy density; $\delta_{I,eq}$ and $\sigma_{I,eq}$ are the equivalent displacement and equivalent stress for a failure mode respectively, and their expressions are given in Ref. [39] in details.

公式 6 和 7 中,进行了损伤状态变量的修正,保证最终计算的 $\delta_{I,eq}^0$ 是正确的,不会出现过大的问题。

$$\begin{aligned} \delta_{I,eq}^0 &= \delta_{I,eq} / \sqrt{F_I} \\ \delta_{I,eq}^f &= 2G_I / \left(\frac{\sigma_{I,eq}}{\sqrt{F_I}} \right) \end{aligned}$$

存在的问题: 什么是 equivalent displacement? $\delta_{I,eq}$, 参考如下^[13]



Table 1

The equivalence displacements and stresses corresponding to different failure modes

Failure modes	Equivalence displacement
$L, \sigma_{11} \geq 0$	$X_{eq}^{Lt} = l \sqrt{(\langle \varepsilon_{11} \rangle)^2 + \alpha \varepsilon_{12}^2 + \alpha \varepsilon_{31}^2}$
$L, \sigma_{11} < 0$	$X_{eq}^{Lc} = l \langle -\varepsilon_{11} \rangle$
$T, \sigma_{22} \geq 0$	$X_{eq}^{Tt} = l \sqrt{(\langle \varepsilon_{22} \rangle)^2 + \alpha \varepsilon_{12}^2 + \alpha \varepsilon_{23}^2}$
$T, \sigma_{22} < 0$	$X_{eq}^{Tc} = l \langle -\varepsilon_{22} \rangle$
$Z, \sigma_{33} \geq 0$	$X_{eq}^{Zt} = l \sqrt{(\langle \varepsilon_{33} \rangle)^2 + \alpha \varepsilon_{23}^2 + \alpha \varepsilon_{31}^2}$
$Z, \sigma_{33} < 0$	$X_{eq}^{Zc} = l \langle -\varepsilon_{33} \rangle$

where l is the characteristic length of element.

问题：运算 $\langle x \rangle = (x + |x|)/2$ 如何理解？

The symbol $\langle \rangle$ in the equations above represents the Macaulay bracket operator
麦考利函数, [anaqus analysis user's guide>materials>24 progress damage and failure>24.3.3 damage evolution and element removal for fiber-reinforced composites]

$$\langle x \rangle = \begin{cases} 0, & x \leq 0 \\ x, & x \geq 0 \end{cases}$$

2.2.3 Abaqus

基于二维 hashin

Damage Initiation



The initiation criteria have the following general forms:

Fiber tension ($\hat{\sigma}_{11} \geq 0$):

$$F_f^t = \left(\frac{\hat{\sigma}_{11}}{X^T}\right)^2 + \alpha \left(\frac{\hat{\tau}_{12}}{S^L}\right)^2.$$

Fiber compression ($\hat{\sigma}_{11} < 0$):

$$F_f^c = \left(\frac{\hat{\sigma}_{11}}{X^C}\right)^2.$$

Matrix tension ($\hat{\sigma}_{22} \geq 0$):

$$F_m^t = \left(\frac{\hat{\sigma}_{22}}{Y^T}\right)^2 + \left(\frac{\hat{\tau}_{12}}{S^L}\right)^2.$$

Matrix compression ($\hat{\sigma}_{22} < 0$):

$$F_m^c = \left(\frac{\hat{\sigma}_{22}}{2S^T}\right)^2 + \left[\left(\frac{Y^C}{2S^T}\right)^2 - 1\right] \frac{\hat{\sigma}_{22}}{Y^C} + \left(\frac{\hat{\tau}_{12}}{S^L}\right)^2.$$

Where,

$$\hat{\sigma}_{11}, \hat{\sigma}_{22}, \hat{\tau}_{12}$$

are components of the effective stress tensor, $\hat{\sigma}$, that is used to evaluate the initiation criteria and which is computed from:

$$\hat{\sigma} = \mathbf{M}\sigma,$$

where σ is the true stress and $\mathbf{M}$ is the damage operator:

$$\mathbf{M} = \begin{bmatrix} \frac{1}{(1-d_f)} & 0 & 0 \\ 0 & \frac{1}{(1-d_m)} & 0 \\ 0 & 0 & \frac{1}{(1-d_s)} \end{bmatrix}.$$

d_f , d_m , and d_s are internal (damage) variables that characterize fiber, matrix, and shear damage, which are derived from damage variables d_f^t , d_f^c , d_m^t , and d_m^c , corresponding to the four modes previously discussed, as follows:

$$\begin{aligned} d_f &= \begin{cases} d_f^t & \text{if } \hat{\sigma}_{11} \geq 0, \\ d_f^c & \text{if } \hat{\sigma}_{11} < 0, \end{cases} \\ d_m &= \begin{cases} d_m^t & \text{if } \hat{\sigma}_{22} \geq 0, \\ d_m^c & \text{if } \hat{\sigma}_{22} < 0, \end{cases} \\ d_s &= 1 - (1 - d_f^t)(1 - d_f^c)(1 - d_m^t)(1 - d_m^c). \end{aligned}$$

Prior to any damage initiation and evolution the damage operator, $\mathbf{M}$, is equal to the identity matrix, so $\hat{\sigma} = \sigma$. Once damage initiation and evolution has occurred for at least one mode, the damage operator becomes significant in the criteria for damage initiation of other modes (see “Damage evolution and element removal for fiber-reinforced composites,” Section 24.3.3, for discussion of damage evolution). The



effective stress, σ , is intended to represent the stress acting over the damaged area that effectively resists the internal forces.

Damage evolution

Prior to damage initiation the material is linearly elastic, with the stiffness matrix of a plane stress orthotropic material. Thereafter, the response of the material is computed from

$$\sigma = \mathbf{C}_d \varepsilon,$$

where ε is the strain and $\mathbf{C}_d$ is the damaged elasticity matrix, which has the form

$$\mathbf{C}_d = \frac{1}{D} \begin{bmatrix} (1-d_f)E_1 & (1-d_f)(1-d_m)\nu_{21}E_1 & 0 \\ (1-d_f)(1-d_m)\nu_{12}E_2 & (1-d_m)E_2 & 0 \\ 0 & 0 & (1-d_s)GD \end{bmatrix},$$

where $D = 1 - (1-d_f)(1-d_m)\nu_{12}\nu_{21}$, d_f reflects the current state of fiber damage, d_m reflects the current state of matrix damage, d_s reflects the current state of shear damage, E_1 is the Young's modulus in the fiber direction, E_2 is the Young's modulus in the matrix direction, G is the shear modulus, and ν_{12} and ν_{21} are Poisson's ratios.

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} \frac{E_1}{1-\nu_{12}\nu_{21}} & \frac{\nu_{21}E_1}{1-\nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} & \frac{E_2}{1-\nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix} \begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{Bmatrix} = \mathbf{Q} \varepsilon$$

其中的 $\mathbf{C}_d$ 如何推导?

The damage variables d_f , d_m , and d_s are derived from damage variables d_f^t , d_f^c , d_m^t , and d_m^c , corresponding to the four failure modes previously discussed, as follows:

$$d_f = \begin{cases} d_f^t & \text{if } \hat{\sigma}_{11} \geq 0, \\ d_f^c & \text{if } \hat{\sigma}_{11} < 0, \end{cases}$$

$$d_m = \begin{cases} d_m^t & \text{if } \hat{\sigma}_{22} \geq 0, \\ d_m^c & \text{if } \hat{\sigma}_{22} < 0, \end{cases}$$

$$d_s = 1 - (1-d_f^t)(1-d_f^c)(1-d_m^t)(1-d_m^c),$$

$\hat{\sigma}_{11}$ and $\hat{\sigma}_{22}$ are components of the effective stress tensor. The effective stress tensor is primarily used to evaluate damage initiation criteria; see [“Damage initiation for fiber-reinforced composites,” Section 24.3.2](#), for a description of how the effective stress tensor is computed.



Equivalent displacement and stress for each of the four damage modes are defined as follows:

Fiber tension ($\hat{\sigma}_{11} \geq 0$):

$$\delta_{eq}^{ft} = L^c \sqrt{\langle \varepsilon_{11} \rangle^2 + \alpha \varepsilon_{12}^2},$$

$$\sigma_{eq}^{ft} = \frac{\langle \sigma_{11} \rangle \langle \varepsilon_{11} \rangle + \alpha \tau_{12} \varepsilon_{12}}{\delta_{eq}^{ft} / L^c},$$

Fiber compression ($\hat{\sigma}_{11} < 0$):

$$\delta_{eq}^{fc} = L^c \langle -\varepsilon_{11} \rangle,$$

$$\sigma_{eq}^{fc} = \frac{\langle -\sigma_{11} \rangle \langle -\varepsilon_{11} \rangle}{\delta_{eq}^{fc} / L^c}.$$

Matrix tension ($\hat{\sigma}_{22} \geq 0$):

$$\delta_{eq}^{mt} = L^c \sqrt{\langle \varepsilon_{22} \rangle^2 + \varepsilon_{12}^2},$$

$$\sigma_{eq}^{mt} = \frac{\langle \sigma_{22} \rangle \langle \varepsilon_{22} \rangle + \tau_{12} \varepsilon_{12}}{\delta_{eq}^{mt} / L^c}.$$

Matrix compression ($\hat{\sigma}_{22} < 0$):

$$\delta_{eq}^{mc} = L^c \sqrt{\langle -\varepsilon_{22} \rangle^2 + \varepsilon_{12}^2},$$

$$\sigma_{eq}^{mc} = \frac{\langle -\sigma_{22} \rangle \langle -\varepsilon_{22} \rangle + \tau_{12} \varepsilon_{12}}{\delta_{eq}^{mc} / L^c}.$$

The characteristic length, L^c , is based on the element geometry and formulation

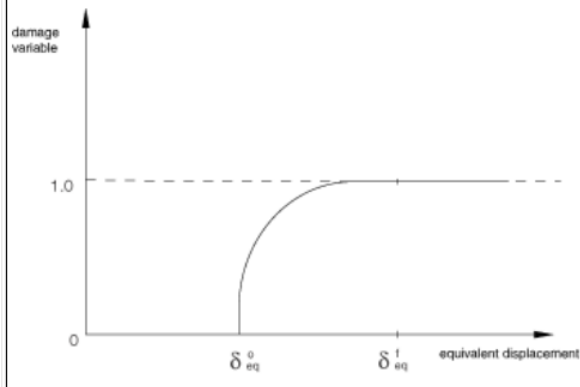
The symbol $\langle \rangle$ in the equations above represents the Macaulay bracket operator

After damage initiation (i.e., $\delta_{eq} \geq \delta_{eq}^0$) for the behavior shown in [Figure 24.3.3-1](#), the damage variable for a particular mode is given by the following expression

$$d = \frac{\delta_{eq}^f (\delta_{eq} - \delta_{eq}^0)}{\delta_{eq} (\delta_{eq}^f - \delta_{eq}^0)},$$

where δ_{eq}^0 is the initial equivalent displacement at which the initiation criterion for that mode was met and δ_{eq}^f is the displacement at which the material is completely damaged in this failure mode. The above relation is presented graphically in [Figure 24.3.3-2](#).



Figure 24.3.3–1 Equivalent stress versus equivalent displacement.**Figure 24.3.3–2** Damage variable as a function of equivalent displacement.

2.2.4 Ref^[6] hashin + equivalent displacement

between the effective stress ($\hat{\sigma}$) and nominal stress (σ) can be defined in form:

$$\hat{\sigma} = d \cdot \sigma \quad (1)$$

where d is the damage operator

Thus stress with presence of the damages is given as:

$$\sigma_{ij} = C_{ij}(d) \cdot \varepsilon_{ij} \quad (2)$$

where C represents the undamaged orthotropic stiffness matrix. This matrix takes the following form:

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{13} & C_{23} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \quad (3)$$

Then the damage stiffness matrix as follow [23]:

$$\begin{aligned} dC_{11} &= (1 - d_f)E_1(1 - \nu_{23}^2)\Gamma, \\ dC_{22} &= (1 - d_f)(1 - d_m)E_2(1 - \nu_{13}^2)\Gamma \\ dC_{33} &= (1 - d_f)(1 - d_m)E_3(1 - \nu_{21}^2)\Gamma, \\ dC_{12} &= (1 - d_f)(1 - d_m)E_1(\nu_{21} - \nu_{31}\nu_{23})\Gamma \\ dC_{23} &= (1 - d_f)(1 - d_m)E_2(\nu_{32} - \nu_{12}\nu_{31})\Gamma, \\ dC_{31} &= (1 - d_f)(1 - d_m)E_1(\nu_{31} - \nu_{21}\nu_{32})\Gamma \\ dC_{44} &= (1 - d_f)(1 - d_{mt}s_{mt})E_1(1 - d_{mc}s_{mc})G_{12}, \\ dC_{55} &= (1 - d_f)(1 - d_{mt}s_{mt})E_1(1 - d_{mc}s_{mc})G_{23} \\ dC_{66} &= (1 - d_f)(1 - d_{mt}s_{mt})E_1(1 - d_{mc}s_{mc})G_{31}, \end{aligned} \quad (4)$$

where the damage variables and are given by Eq. (5) [23].

$$\begin{aligned} d_f &= 1 - (1 - d_{ft})(1 - d_{fc}) \\ d_m &= 1 - (1 - d_{mt})(1 - d_{mc}) \\ \Gamma &= 1/(1 - \nu_{12}^2 - \nu_{23}^2 - \nu_{13}^2 - 2\nu_{12}\nu_{23}\nu_{13}) \end{aligned} \quad (5)$$

其中的泊松比错误，比如第一行中的 ν_{23}^2 ，应该修改为 $\nu_{23}\nu_{32}$

where d_f , d_m and d_s is the damage variables for the fiber, matrix and shear failure mode respectively.

其中的 $s_{mt}=0.9$, $s_{mc}=0.5$

Damage variables:



As previously specified, after the initial failure criteria have been satisfied, and based on the relationship between the effective stress and displacement, the damage variable for each mode in fiber and matrix can be illustrated in form:

$$d_i = \frac{\delta_{i,eq}^f (\delta_{i,eq} - \delta_{i,eq}^0)}{\delta_{i,eq}^f (\delta_{i,eq}^f - \delta_{i,eq}^0)}, \quad \delta_{i,eq}^0 \leq \delta_{i,eq} \leq \delta_{i,eq}^f \quad (6)$$

where d_i is the damage variable defined for each damage mode. The variables δ_{eq}^0 , δ_{eq} and $\delta_{i,eq}^f$ represent respectively the initial displacement, equivalent displacement and equivalent failure displacement which are computed as follow:

where d_i is the damage variable defined for each damage mode. The variables δ_{eq}^0 , δ_{eq} and $\delta_{i,eq}^f$ represent respectively the initial displacement, equivalent displacement and equivalent failure displacement which are computed as follow:

$$\begin{aligned} G_i &= \int_{\delta_{i,eq}^0}^{\delta_{i,eq}^f} \sigma d\epsilon = \frac{1}{2} (\delta_{i,eq}^f - \delta_{i,eq}^0) \sigma_{i,eq}^0 \\ \delta_{i,eq}^f - \delta_{i,eq}^0 &= \frac{2G_i}{\sigma_{i,eq}^0} \\ \delta_{i,eq}^0 &= \sigma_{i,eq} / \sqrt{f_i} \\ \sigma_{i,eq}^0 &= \sigma_{i,eq} / \sqrt{f_i} \end{aligned} \quad (7)$$

where $\sigma_{i,eq}$, $\sigma_{i,eq}^0$, f_i and G_i are the equivalent stress, equivalent initial stress, initial failure and the fracture energy corresponding for each mode, respectively. These variables are computed at each element's integration point to describe the materials stiffness degradation. This scheme is shown in Fig. 1.



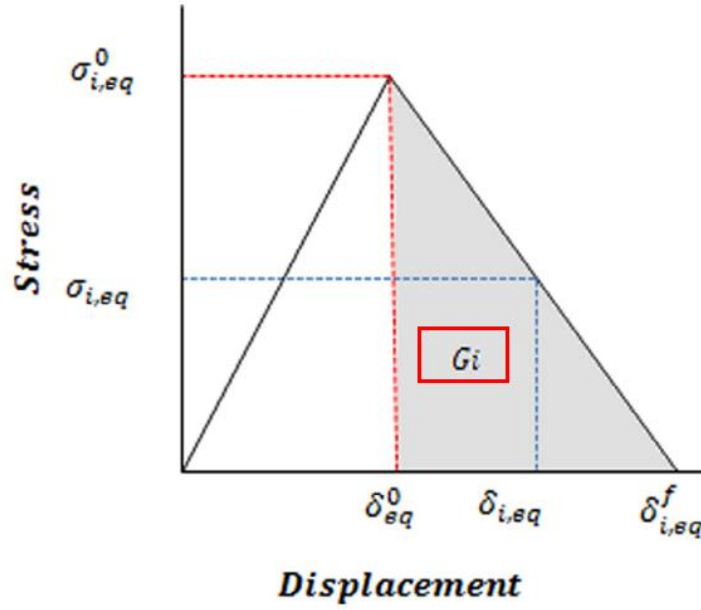


Fig. 1. Typical behavior of material stiffness degradation.

其中的 fracture energy 计算不同，应该为整体曲线下的面积，如下：

$$G_i = \int_0^{\delta_{i,eq}^f} \sigma d\varepsilon = \frac{1}{2} \delta_{i,eq}^f * \sigma_{i,eq}^0$$

Damage Initiation

Table 1
Hashin failure criteria.

Fiber Tensile Failure ($\sigma_{11} \geq 0$)	$f_{ft} = \left(\frac{\sigma_{11}}{X_t}\right)^2 \geq 1$
Fiber Compression Failure ($\sigma_{11} < 0$)	$f_{fc} = \left(\frac{\sigma_{11}}{X_c}\right)^2 \geq 1$
Matrix Tensile Failure ($\sigma_{22} + \sigma_{33} \geq 0$)	$f_{mt} = \frac{(\sigma_{22} + \sigma_{33})^2}{Y_t^2} + \frac{\sigma_{23}^2 - \sigma_{22}\sigma_{33}}{S_{23}^2} + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1$
Matrix Compression Failure ($\sigma_{22} + \sigma_{33} < 0$)	$f_{mc} = \frac{1}{Y_c} \left(\left(\frac{Y_c}{2S_{23}} \right)^2 - 1 \right) (\sigma_{22} + \sigma_{33}) + \frac{(\sigma_{22} + \sigma_{33})^2}{4S_{23}^2} + \frac{\sigma_{23}^2 - \sigma_{22}\sigma_{33}}{S_{23}^2} + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2 \geq 1$

2.2.5 Ref^[7] 二维 hashin + equivalent displacement

整体逻辑清晰，可以参考逻辑



sors and damage evolution laws. Based on the CDM theory, the damage constitutive model for the in-plane damage of composite laminates can be expressed as [26]:

$$C_d = \frac{1}{D} \begin{bmatrix} (1-d_f)E_1 & (1-d_f)(1-d_m)v_{21}E_1 & 0 \\ (1-d_f)(1-d_m)v_{12}E_2 & (1-d_m)E_2 & 0 \\ 0 & 0 & (1-d_s)GD \end{bmatrix} \quad (1)$$

$$D = 1 - (1-d_f)(1-d_m)v_{12}v_{21} \quad (2)$$

where d_f , d_m and d_s are damage variables range from 0 to 1, representing the fiber breakage, matrix cracking and shear failure respectively. In addition, the shear failure is not independent which can be considered as the interaction of fiber breakage and matrix cracking failure. Thus the damage variable relating to shear is given by:

$$d_s = 1 - (1-d_f^t)(1-d_m^c)(1-d_f^t)(1-d_m^c)$$

Damage initiation

structure subjected to internal pressure. Thus, fiber dominated damage initiation is determined using a simple non-interacting Max-stress criterion:

$$F_f^t = \frac{\sigma_{11}}{X^T} = 1 \text{ Fiber tension } (\sigma_{11} \geq 0) \quad (4)$$

where X^T is ply tensile strength in the fiber direction.

The Hashin failure criterion is used to determine the initiation of matrix damage, which can identify the detailed failure modes, i.e. matrix tension and compression failure. Besides, the Hashin criterion considers the contribution of longitudinal shear stress to each failure mode:

$$F_m^t = \left(\frac{\sigma_{22}}{Y^T} \right)^2 + \left(\frac{\tau_{12}}{S^t} \right)^2 = 1 \text{ Matrix tension } (\sigma_{22} \geq 0) \quad (5)$$

$$F_m^c = \left(\frac{\sigma_{22}}{2S^T} \right)^2 + \left[\left(\frac{Y^c}{2S^T} \right)^2 - 1 \right] \frac{\sigma_{22}}{Y^c} + \left(\frac{\tau_{12}}{S^t} \right)^2 = 1 \text{ Matrix compression } (\sigma_{22} < 0) \quad (6)$$

Damage evolution



Once the above failure criterion is satisfied, material stiffness of the damaged element will degrade gradually based on a linear softening law, which the strength of the ply decreases linearly with strain as shown in Fig. 1. This degradation law is determined based on the assumption that the total energy needed to fail an element equals to the energy needed to create a crack that passes through it. Thus the fracture energy can be expressed as a function of the equivalent stress and equivalent displacement:

$$\int_0^{\infty} \sigma_{eq} d(\delta_{eq}) = G_c \quad (7)$$

where δ_{eq} is the equivalent displacement defined by the corresponding strain components multiplied by the characteristic element length l_c ; σ_{eq} is the equivalent stress for the same failure mode, which can be obtained by relating the equivalent displacement to the elastic deformation energy:

For fiber tension:

$$\delta_{eq} = l_c \langle \varepsilon_{11} \rangle, \sigma_{eq} = l_c [\langle \sigma_{11} \rangle \langle \varepsilon_{11} \rangle] / \delta_{eq} \quad (8)$$

For matrix tension:

$$\delta_{eq} = l_c \sqrt{\langle \varepsilon_{22} \rangle^2 + \varepsilon_{12}^2}, \sigma_{eq} = l_c [\langle \sigma_{22} \rangle \langle \varepsilon_{22} \rangle + \tau_{12} \varepsilon_{12}] / \delta_{eq} \quad (9)$$

For matrix compression:

$$\delta_{eq} = l_c \sqrt{\langle -\varepsilon_{22} \rangle^2 + \varepsilon_{12}^2}, \sigma_{eq} = l_c [\langle -\sigma_{22} \rangle \langle -\varepsilon_{22} \rangle + \tau_{12} \varepsilon_{12}] / \delta_{eq} \quad (10)$$

l_c is the characteristic length of the element, which is defined as the square root of the area of a continuum shell element In ABAQUS

damage variables

2.1.4. Damage variables computation

Based on the linear softening law, the damage variable for each failure mode is defined by the equivalent displacement, and can be calculated as the followings:

$$d = 1 - \frac{\delta_{eq}^0 (\delta_{eq}^f - \delta_{eq})}{\delta_{eq} (\delta_{eq}^f - \delta_{eq}^0)}; \delta_{eq}^0 \leq \delta_{eq} \leq \delta_{eq}^f \quad (11)$$

where δ_{eq}^0 is the equivalent displacement when the damage initiates, and δ_{eq}^f is the equivalent displacement at which the material

2.2.6 Ref^[10] 指数 equivalent strain

σ shows the same stress strain response as a comparable undamaged volume under the effective stress, σ^e .



$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \tau_{23} \\ \tau_{13} \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} (1-d_f^t)(1-d_f^c) & 0 & 0 & 0 & 0 & 0 \\ 0 & (1-d_m^t) & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & (1-d_m^t) & 0 & 0 \\ 0 & 0 & 0 & 0 & (1-d_m^t) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma_{11}^e \\ \sigma_{22}^D \\ \sigma_{33}^D \\ \tau_{23}^D \\ \tau_{13}^D \\ \tau_{12}^D \end{bmatrix}$$

考虑了泊松比的影响

To make **C** positive definite, Poisson's ratio is changed to $\nu_{ij} = \nu_{ij}^0(1 - D_{ij})$ whereas ν_{ij}^0 is Poisson's ratio of damage free material.

$$d_f^i = 1 - \exp\left(\frac{1}{me}\left(1 - \frac{\varepsilon_{11}}{\varepsilon_{11}^0}\right)^m\right) \quad (31)$$

where $i = t, c$ for fiber tension and compression respectively. Parameter 'm' is called material softening parameter whereas 'e' is Napier's constant. The material shows brittle behavior for higher values of 'm' and ductile behavior for lower values of 'm'.

In case of linear stiffness decay model, the damage variable is related to strain as

$$d_f^i = \left(\frac{\varepsilon_{11}^{fi}}{\varepsilon_{11}^{fi} - \varepsilon_{11}^0}\right)\left(\frac{\varepsilon_{11} - \varepsilon_{11}^0}{\varepsilon_{11}}\right)$$

where $i = t, c$ for fiber tension and compression respectively (see Fig. 9). ε_{11}^f is the strain corresponding to fully damaged state and signifies a complete loss of stiffness. ε_{11}^0 is the strain corresponding to damage initiation state (see Fig. 2).



3.2.2. Matrix damage evolution

In case of exponential stiffness decay model, the damage variable is related to strain as

$$d_m^t = 1 - \exp\left(\frac{1}{me}\left(1 - \frac{\varepsilon_{22}}{\varepsilon_{22}^0}\right)^m\right) \quad (32)$$

Damage variable in case of matrix compression is related to strain as

$$d_m^c = 1 - \exp\left(\frac{1}{me}\left(1 - \frac{\gamma_r}{\gamma_r^0}\right)^m\right) \quad (33)$$

where

$$\gamma_r = \sqrt{(\gamma_t^n)^2 + (\gamma_l^n)^2} \quad (34)$$

and γ_r is the resultant shear strain on the plane of fracture.

2.2.7 Ref^[14] hashin + equivalent disp

For fiber tension failure ($\sigma_{11} > 0$),

$$e_{ft} = \left(\frac{\sigma_{11}}{F_{1t}}\right)^2 + \alpha \left(\frac{\sigma_{12} + \sigma_{31}}{F_{ls}}\right)^2 \geq 1 \quad (2)$$

For fiber compression failure ($\sigma_{11} < 0$),

$$e_{fc} = \left(\frac{\sigma_{11}}{F_{1c}}\right)^2 \geq 1 \quad (3)$$

For matrix tension failure ($\sigma_{22} > 0$),

$$e_{mt} = \left(\frac{\sigma_{22}}{F_{2t}}\right)^2 + \left(\frac{\sigma_{12}}{F_{ls}}\right)^2 + \left(\frac{\sigma_{23}}{F_{ts}}\right)^2 \geq 1 \quad (4)$$

For matrix compression failure ($\sigma_{22} < 0$),

$$e_{mc} = \frac{1}{4} \left(\frac{-\sigma_{22}}{F_{ls}}\right)^2 + \frac{F_{2c}^2 \sigma_{22}}{4F_{ls}^2 F_{2c}} - \frac{\sigma_{22}}{F_{2c}} + \left(\frac{\sigma_{12}}{F_{ts}}\right)^2 \geq 1 \quad (5)$$

In the above equations, e_{ft} , e_{fc} , e_{mt} , e_{mc} , e_{dd} are damage variables; and F_{1t} , F_{1c} , F_{2t} , F_{2c} , F_{ls} , F_{ts} are axial tensile strength, axial compressive strength, transverse tensile strength, transverse compressive strength, longitudinal shear strength and transverse shear strength of the material, respectively. The through-thickness damage of a fiber bundle is not considered in this work, since there is no through-thickness damage observed in the fiber bundles [4,5].



respectively. The evolution of damage variable for each failure mode I is expressed by the following equation:

$$d_I = \frac{\delta_{I,eq}^f (\delta_{I,eq} - \delta_{I,eq}^0)}{\delta_{I,eq} (\delta_{I,eq}^f - \delta_{I,eq}^0)}, \quad I = ft, fc, mt, \text{ and } mc \quad (7)$$

In the above equation, $\delta_{I,eq}^f$ is the full damage equivalent displacement of the corresponding failure mode, and $\delta_{I,eq}^0$ is the equivalent displacement at which the failure criterion is satisfied. The equivalent displacements and stresses for each failure mode are computed using formulas listed in Table 1. $\delta_{I,eq}^f$ and $\delta_{I,eq}^0$ can be computed by the following equations:

$$\delta_{I,eq}^f = \frac{2G_I}{\sigma_{I,eq}^0} \quad (8)$$

$$\delta_{I,eq}^0 = \frac{\delta_{I,eq}}{\sqrt{e_I}} \quad (9)$$

Here, $\delta_{I,eq}^0$ and $\sigma_{I,eq}^0$ denote the initiation damage equivalence displacement and stress, respectively. e_I is the value of damage criteria.

Table 1
Equivalent displacement and stress of each failure mode.

Failure modes	Equivalence displacement	Equivalence stress
Fiber tension	$\delta_{ft,eq} = l_c \sqrt{(\langle \varepsilon_{11} \rangle)^2 + \alpha \varepsilon_{12}^2 + \alpha \varepsilon_{31}^2}$	$\sigma_{ft,eq} = \frac{l_c (\langle \sigma_{11} \rangle \langle \varepsilon_{11} \rangle + \alpha \sigma_{12} \varepsilon_{12} + \alpha \sigma_{31} \varepsilon_{31})}{\delta_{ft,eq}}$
Fiber compression	$\delta_{fc,eq} = l_c \langle -\varepsilon_{11} \rangle$	$\sigma_{fc,eq} = \frac{l_c \langle -\sigma_{11} \rangle \langle -\varepsilon_{11} \rangle}{\delta_{fc,eq}}$
Matrix tension	$\delta_{mt,eq} = l_c \sqrt{(\langle \varepsilon_{22} \rangle)^2 + \varepsilon_{12}^2 + \varepsilon_{23}^2}$	$\sigma_{mt,eq} = \frac{l_c (\langle \sigma_{22} \rangle \langle \varepsilon_{22} \rangle + \sigma_{12} \varepsilon_{12} + \sigma_{23} \varepsilon_{23})}{\delta_{mt,eq}}$
Matrix compression	$\delta_{mc,eq} = l_c \sqrt{(\langle -\varepsilon_{22} \rangle)^2 + \varepsilon_{12}^2}$	$\sigma_{mc,eq} = \frac{l_c (\langle -\sigma_{22} \rangle \langle -\varepsilon_{22} \rangle + \sigma_{12} \varepsilon_{12})}{\delta_{mc,eq}}$



And the corresponding damaged stiffness matrix $C(d)$ is obtained as following:

$$C(d) = \frac{1}{\Delta} \begin{bmatrix} d_f E_{11}(1 - d_m v_{23} v_{32}) & d_f d_m E_{11}(v_{21} + v_{23} v_{31}) & d_f E_{11}(v_{31} + v_{21} v_{32}) & & & \\ & d_m E_{22}(1 - d_f v_{13} v_{31}) & d_m E_{22}(v_{32} + d_f v_{12} v_{31}) & & & \\ & & E_{33}(1 - d_f d_m v_{12} v_{21}) & & & \\ & & & d_f d_m G_{12} & & \\ & & & d_m G_{23} & & \\ & & & & d_f G_{13} & \end{bmatrix} \quad (11)$$

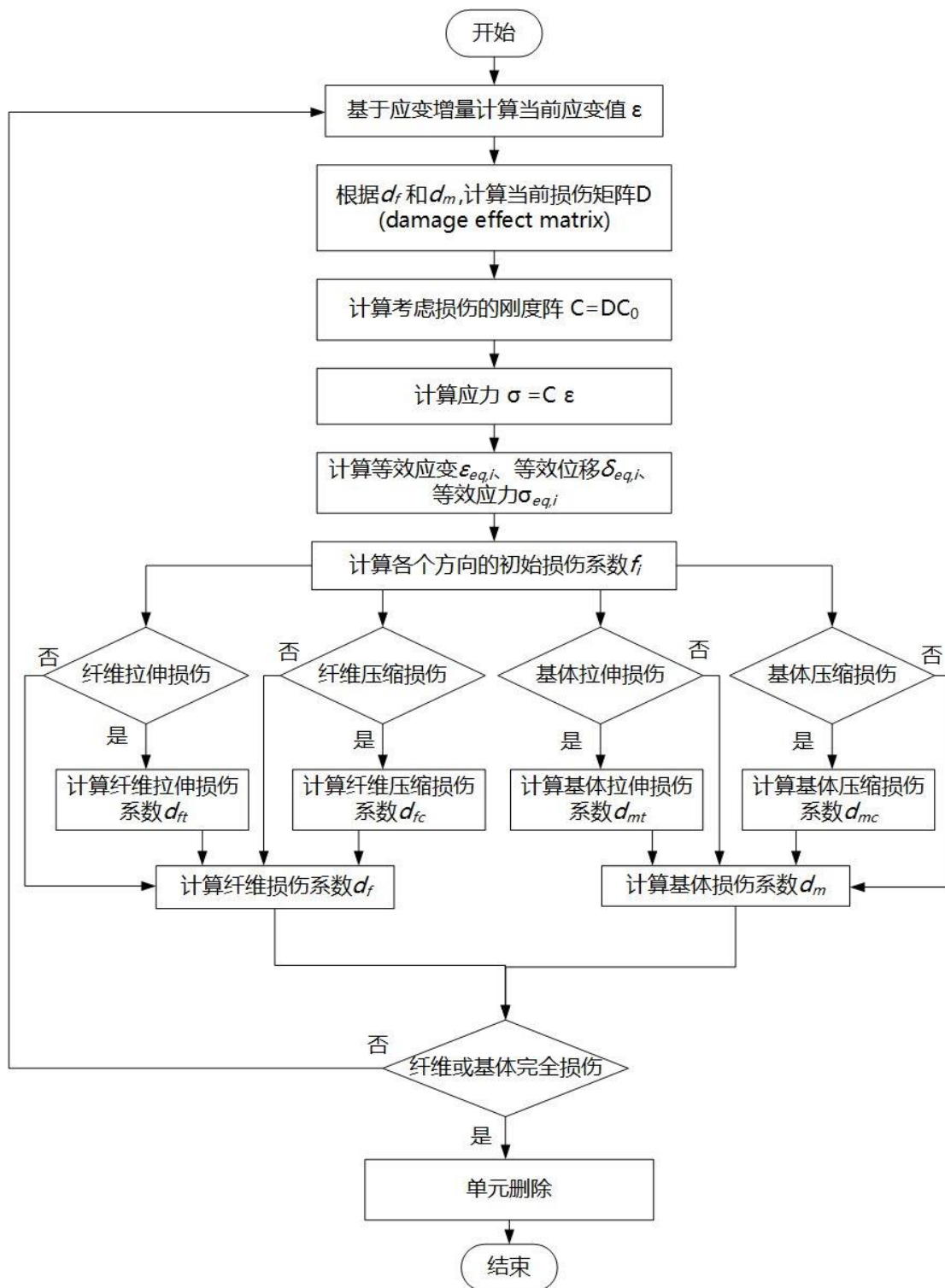
where $d_f = (1 - d_{ft})(1 - d_{fc})$, $d_m = (1 - d_{mt})(1 - d_{mc})$

$$\Delta = \frac{1}{1 - d_f d_m v_{12} v_{21} - d_m v_{23} v_{32} - d_f v_{13} v_{31} - 2d_f d_m v_{21} v_{32} v_{13}}$$

d_f and d_m are global damage variables associated with fiber failure and matrix failure, respectively, while, d_{ft} , d_{fc} , d_{mt} , and d_{mc} are the damage-evolution variables associated with the four damage modes: fiber tension, fiber compression, matrix tension and matrix compression.



3 最终使用的材料模型



3.1 弹性阶段

单向纤维增强复合材料属于正交各向复合材料，为了更好的考虑面外位移，本文采用了三维本构模型，应力应变关系如下：

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{bmatrix} = \begin{bmatrix} E_{11}(1-\nu_{23}\nu_{32})\Gamma & E_{11}(\nu_{21}+\nu_{31}\nu_{23})\Gamma & E_{11}(\nu_{31}+\nu_{21}\nu_{32})\Gamma & 0 & 0 & 0 \\ E_{11}(\nu_{21}+\nu_{31}\nu_{23})\Gamma & E_{22}(1-\nu_{13}\nu_{31})\Gamma & E_{22}(\nu_{32}+\nu_{12}\nu_{31})\Gamma & 0 & 0 & 0 \\ E_{11}(\nu_{31}+\nu_{21}\nu_{32})\Gamma & E_{22}(\nu_{32}+\nu_{12}\nu_{31})\Gamma & E_{33}(1-\nu_{12}\nu_{21})\Gamma & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{23} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{13} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{31} \end{bmatrix}$$

其中， σ_{ij} ($i,j=1,2,3$)为应力分量， ε_{ij} ($i,j=1,2,3$) 为应变分量， E_{ii} ($i=1,2,3$) 为拉伸模量， G_{ij} ($i,j=1,2,3, i \neq j$)为剪切模量， ν_{ij} ($i,j=1,2,3, i \neq j$) 为泊松比，1、2、3 分别代表纤维方向、面内垂直方向以及面外垂直方向。 Γ 定义如下：

$$\Gamma = 1 / (1 - \nu_{12}\nu_{21} - \nu_{23}\nu_{32} - \nu_{13}\nu_{31} - 2\nu_{21}\nu_{32}\nu_{13})$$

并且，泊松比与弹性模量满足下式^[5]：

$$\frac{\nu_{12}}{E_{11}} = \frac{\nu_{21}}{E_{22}}, \frac{\nu_{13}}{E_{11}} = \frac{\nu_{31}}{E_{33}}, \frac{\nu_{23}}{E_{22}} = \frac{\nu_{32}}{E_{33}}$$

3.2 Damage effect tensor(matrix) D [6]

σ shows the same stress strain response as a comparable undamaged volume under the effective stress, σ^e is the effective stress tensor considering the damage.

$$\sigma^e = C\sigma = DC_0\varepsilon$$

值得注意的是，不同文献中 D(Damage effect matrix)的表达式不同。

$$\begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{31} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{13} & C_{23} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 2G_{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 2G_{23} & 0 \\ 0 & 0 & 0 & 0 & 0 & 2G_{31} \end{pmatrix} \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{31} \end{pmatrix}$$



$$\begin{aligned}
C_{11} &= (1 - d_f)C_{11}^0 \\
C_{22} &= (1 - d_f)(1 - d_m)C_{22}^0 \\
C_{33} &= (1 - d_f)(1 - d_m)C_{33}^0 \\
C_{12} &= (1 - d_f)(1 - d_m)C_{12}^0 \\
C_{23} &= (1 - d_f)(1 - d_m)C_{23}^0 \\
C_{13} &= (1 - d_f)(1 - d_m)C_{13}^0 \\
G_{12} &= (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{12}^0 \\
G_{23} &= (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{23}^0 \\
G_{31} &= (1 - d_f)(1 - s_{mt}d_{mt})(1 - s_{mc}d_{mc})G_{31}^0
\end{aligned}$$

Damage variables:

$$\begin{aligned}
d_f &= 1 - (1 - d_{ft})(1 - d_{fc}) \\
d_m &= 1 - (1 - d_{mt})(1 - d_{mc})
\end{aligned}$$

The last three equations include the factors to aide in controlling the shear stiffness loss due to failure of the matrix in tension, s_{mt} , and in compression, s_{mc} , The factors were assumed to be 0.9 and 0.5, respectively.

注意所有损伤系数 damage value 不可逆

damaged values d_{ij}

$$\begin{aligned}
d &= \max(d(t + \Delta t), d(t), 0) \\
d &= [0, 1]
\end{aligned}$$

To account for irreversibility, the damage variable as a function of analysis time, t , defined as

$$\begin{cases} d_{ii}(t + \Delta t) = \max\{0, \min\{1, d_{ii}(t + \Delta t)\}\} \\ d_{ii}(t + \Delta t) \geq d_{ii}(t) \end{cases}, i = 1, 2, 3. \quad (4)$$

3.3 Damage initiation

不同使用工况下，三维 Hshin 准则的表达式存在一定差异，本文使用下列表达式，如下：

纤维拉伸 ($\sigma_{11} \geq 0$):

$$f_{ft} = \left(\frac{\sigma_{11}}{S_{11T}}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2$$

纤维压缩 ($\sigma_{11} < 0$):

$$f_{fc} = \left(\frac{\sigma_{11}}{S_{11C}}\right)^2$$



基体拉伸 ($\sigma_{22} + \sigma_{33} \geq 0$):

$$f_{mt} = \left(\frac{\sigma_{22} + \sigma_{33}}{S_{22T}} \right)^2 + \left(\frac{\sigma_{12}}{S_{12}} \right)^2 + \left(\frac{\sigma_{13}}{S_{13}} \right)^2 + \frac{\sigma_{23}^2 - \sigma_{22}\sigma_{33}}{S_{23}}$$

基体压缩 ($\sigma_{22} + \sigma_{33} < 0$):

$$f_{mc} = \left(\frac{\sigma_{22} + \sigma_{33}}{S_{22C}} \right)^2$$

其中, 1、2、3 分别代表纤维方向 (X)、面内垂直方向 (Y) 以及面外垂直方向 (Z), 如图 2.1 所示。 S_{11T} 和 ϵ_{11C} 为纤维方向的拉伸与压缩失效应变, S_{22T} 和 S_{22C} 为基体方向的拉伸和压缩失效应变, $S_{ijf} (i,j=1,2,3, i \neq j)$ 为剪切失效应变。复合材料层可以同时存在多种损伤模式, 当任何一个损伤参数 ($f_{ft}, f_{fc}, f_{mt}, f_{mc}$) 达到 1 时, 该单元则满足了对应的损伤初始准则, 而后发生刚度退化。

不考虑剪切损伤, 因为没有找到剪切的 **damage variables**。在程序中加入了面内和面外的初始损伤判据, 如下, 目的是为了后续增加剪切的损伤演化。如果有剪切的损伤演化需求, 那么可以增加剪切的损伤演化。

面内剪切 (1-2 平面):

$$f_{ip} = \left(\frac{\sigma_{12}}{S_{12}} \right)^2$$

面外剪切 (1-3 与 2-3 平面):

$$f_{op} = \left(\frac{\sigma_{13}}{S_{13}} + \frac{\sigma_{23}}{S_{23}} \right)^2$$

3.4 Damage evolution

当材料满足损伤初始准则时, 即判定该单元开始发生损伤, 材料失去承载能力并且发生刚度退化。在宏观上, 表现为材料的弹性模量发生折减。如果单元发生基体失效或者纤维失效这个单元则被删除。

3.4.1 几个关键问题:

如何计算 **damage variables**, 使用 **equivalent displacement or equivalent strain**.

如何计算 **equivalent displacement or equivalent strain**?

如何计算特征长度?

如何计算 **equivalent displacement/stain for damage initiation**?



3.4.2 equivalent strain -based damage variables^{[9] [10] [11]}

2.1.3.1. Equivalent strain method.

$$d_i = \frac{\varepsilon_{eq,i}^f (\varepsilon_{eq,i} - \varepsilon_{eq,i}^0)}{\varepsilon_{eq,i}^f (\varepsilon_{eq,i}^f - \varepsilon_{eq,i}^0)} \quad (11)$$

with $\varepsilon_{eq,i}^0 = X_i/E_{0,j}$, $\varepsilon_{eq,i}^f = 2G_i/(X_i l_c)$, where $\varepsilon_{eq,i}^0$ and $\varepsilon_{eq,i}^f$ represent the initial and final failure strain, respectively; X_i is the fiber or matrix strength; $E_{0,j}$ is the initial modulus of the composite lamina in the corresponding direction; G_i is the critical fracture energy for the specific damage mode; and l_c is the characteristic length of the element, which is introduced to eliminate the mesh sensitivity. In the present study, l_c is considered to be equal to the cube root of the element volume.

3.4.3 equivalent displacement -based damage variables^{[9] [3]}

[6]

2.1.3.2. Equivalent displacement method. The equivalent displacement method can be described by the following equation.

$$d_i = \frac{\delta_{eq,i}^f (\delta_{eq,i} - \delta_{eq,i}^0)}{\delta_{eq,i}^f (\delta_{eq,i}^f - \delta_{eq,i}^0)} \quad (12)$$

in which $\delta_{eq,i}^0 = \delta_{eq,i}/\sqrt{r_i}$ and $\delta_{eq,i}^f = 2G_i/\sigma_{eq,i}^0$, where $\delta_{eq,i}^0$ and $\delta_{eq,i}^f$ denote the initial and fully damaged equivalent displacements, respectively; r_i is the value of the corresponding failure criterion; G_i is the critical fracture energy for the specific damage mode, the same as in Eq. (11); and $\sigma_{eq,i}^0$ is the equivalent stress at the instant of damage initiation. Detailed expressions for calculating the equivalent displacement and equivalent stress for each failure mode can be found in Ref. [33].

3.4.4 equivalent displacement/strain for damage initiation

修正的两种方法如下:

方法 1 ^{[11] [12]}: damage variable 在[0,1]范围内变化, 同时还需要保证损伤不可逆



To illustrate the irreversibility of the damage, the damage variable is defined by the following function of analysis time t and incremental time Δt ,

$$d_{ii}(t + \Delta t) = \max\{0, \min\{1, d_{ii}(t + \Delta t)\}\}, i = 1, 2, 3. \quad (4)$$

使得 damage variable 在 $[0,1]$ 范围内变化, 同时还需要保证损伤不可逆^[12],

To account for irreversibility, the damage variable as a function of analysis time, t , defined as

$$\begin{cases} d_{ii}(t + \Delta t) = \max\{0, \min\{1, d_{ii}(t + \Delta t)\}\} \\ d_{ii}(t + \Delta t) \geq d_{ii}(t) \end{cases}, i = 1, 2, 3. \quad (4)$$

方法 2: 计算 equivalent displacement/stain for damage initiation 时, 除以初始损伤的系数

参考^[3]

where $\delta_{l,eq}^0$ is the equivalent displacement for damage initiation; $\delta_{l,eq}^f$ is the equivalent displacement at final failure; and they can be determined by

$$\delta_{l,eq}^0 = \delta_{l,eq} / \sqrt{F_1} \quad (6)$$

$$\delta_{l,eq}^f = 2G_l / (\sigma_{l,eq} / \sqrt{F_1}) \quad (7)$$

where F_1 is the value of damage initiation criterion; G_l is the fracture energy density; $\delta_{l,eq}$ and $\sigma_{l,eq}$ are the equivalent displacement and equivalent stress for a failure mode respectively, and their expressions are given in Ref. [39] in details.

参考^[6]

$$\delta_{i,eq}^0 = \delta_{i,eq} / \sqrt{f_i}$$

$$\sigma_{i,eq}^0 = \sigma_{i,eq} / \sqrt{f_i}$$

where $\sigma_{i,eq}$, $\sigma_{i,eq}^0$, f_i and G_i are the equivalent stress, equivalent initial stress, initial failure and the fracture energy corresponding for each mode, respectively. These variables are computed at each element's integration point to describe the materials stiffness degradation. This scheme is shown in Fig. 1.

3.4.5 计算 initial and final damaged strain 的方式:

方法 1: ^[9] 直接使用强度



2.1.3.1. Equivalent strain method.

$$d_i = \frac{\varepsilon_{eq,i}^f (\varepsilon_{eq,i} - \varepsilon_{eq,i}^0)}{\varepsilon_{eq,i} (\varepsilon_{eq,i}^f - \varepsilon_{eq,i}^0)} \quad (11)$$

with $\varepsilon_{eq,i}^0 = X_i/E_{0,j}$, $\varepsilon_{eq,i}^f = 2G_i/(X_i l_c)$, where $\varepsilon_{eq,i}^0$ and $\varepsilon_{eq,i}^f$ represent the initial and final failure strain, respectively; X_i is the fiber or matrix strength; $E_{0,j}$ is the initial modulus of the composite lamina in the corresponding direction; G_i is the critical fracture energy for the specific damage mode; and l_c is the characteristic length of the element, which is introduced to eliminate the mesh sensitivity. In the present study, l_c is considered to be equal to the cube root of the element volume.

方法 2: 除以初始损伤的系数
参考^[3]

where $\delta_{l,eq}^0$ is the equivalent displacement for damage initiation; $\delta_{l,eq}^f$ is the equivalent displacement at final failure; and they can be determined by

$$\delta_{l,eq}^0 = \delta_{l,eq} / \sqrt{F_I} \quad (6)$$

$$\delta_{l,eq}^f = 2G_I / (\sigma_{l,eq} / \sqrt{F_I}) \quad (7)$$

where F_I is the value of damage initiation criterion; G_I is the fracture energy density; $\delta_{l,eq}$ and $\sigma_{l,eq}$ are the equivalent displacement and equivalent stress for a failure mode respectively, and their expressions are given in Ref. [39] in details.

参考^[6]

$$\begin{aligned} \delta_{i,eq}^0 &= \delta_{i,eq} / \sqrt{f_i} \\ \sigma_{i,eq}^0 &= \sigma_{i,eq} / \sqrt{f_i} \end{aligned}$$

where $\sigma_{i,eq}$, $\sigma_{i,eq}^0$, f_i and G_i are the equivalent stress, equivalent initial stress, initial failure and the fracture energy corresponding for each mode, respectively. These variables are computed at each element's integration point to describe the materials stiffness degradation. This scheme is shown in Fig. 1.

3.4.6 计算 equivalent displacement or equivalent strain

参考^[13]



Table 1

The equivalence displacements and stresses corresponding to different failure modes

Failure modes	Equivalence displacement
$L, \sigma_{11} \geq 0$	$X_{eq}^{Lt} = l \sqrt{(\langle \varepsilon_{11} \rangle)^2 + \alpha \varepsilon_{12}^2 + \alpha \varepsilon_{31}^2}$
$L, \sigma_{11} < 0$	$X_{eq}^{Lc} = l \langle -\varepsilon_{11} \rangle$
$T, \sigma_{22} \geq 0$	$X_{eq}^{Tt} = l \sqrt{(\langle \varepsilon_{22} \rangle)^2 + \alpha \varepsilon_{12}^2 + \alpha \varepsilon_{23}^2}$
$T, \sigma_{22} < 0$	$X_{eq}^{Tc} = l \langle -\varepsilon_{22} \rangle$
$Z, \sigma_{33} \geq 0$	$X_{eq}^{Zt} = l \sqrt{(\langle \varepsilon_{33} \rangle)^2 + \alpha \varepsilon_{23}^2 + \alpha \varepsilon_{31}^2}$
$Z, \sigma_{33} < 0$	$X_{eq}^{Zc} = l \langle -\varepsilon_{33} \rangle$

Table 1

The equivalence disp

Failure modes	Equivalence stress
$L, \sigma_{11} \geq 0$	$l(\langle \sigma_{11} \rangle \langle \varepsilon_{11} \rangle + \alpha \sigma_{12} \varepsilon_{12} + \alpha \sigma_{13} \varepsilon_{13}) / X_{eq}^{Lt}$
$L, \sigma_{11} < 0$	$l \langle -\sigma_{11} \rangle \langle -\varepsilon_{11} \rangle / X_{eq}^{Lc}$
$T, \sigma_{22} \geq 0$	$l(\langle \sigma_{22} \rangle \langle \varepsilon_{22} \rangle + \alpha \sigma_{12} \varepsilon_{12} + \alpha \sigma_{23} \varepsilon_{23}) / X_{eq}^{Tt}$
$T, \sigma_{22} < 0$	$l \langle -\sigma_{22} \rangle \langle -\varepsilon_{22} \rangle / X_{eq}^{Tc}$
$Z, \sigma_{33} \geq 0$	$l(\langle \sigma_{33} \rangle \langle \varepsilon_{33} \rangle + \alpha \sigma_{23} \varepsilon_{23} + \alpha \sigma_{31} \varepsilon_{31}) / X_{eq}^{Zt}$
$Z, \sigma_{33} < 0$	$l \langle -\sigma_{33} \rangle \langle -\varepsilon_{33} \rangle / X_{eq}^{Zc}$

$$d_f = 1 - (1 - d_{fi})(1 - d_{fc}) \quad (9)$$

$$d_m = 1 - (1 - d_{mt})(1 - d_{mc}) \quad (10)$$

3.4.7 dm 的计算方法

问题: dm 与 dmt, dmc 有关。dmc 和 dmt 通过下面公式计算,但是存在两个方向的 dmt 和 dmc, 即 22 方向与 33 方向。将 22 方向与 33 方向的损伤系数 damage values 记为 dmt_22, dmc_22, dmt_33, dmc_33。

$$d_i = \frac{\delta_{eq,i}^f (\delta_{eq,i} - \delta_{eq,i}^0)}{\delta_{eq,i}^f (\delta_{eq,i}^f - \delta_{eq,i}^0)}$$

但是 dm 参考文献中的计算公式中,并没有考虑到 22 和 33 方向。也没有找到相关的文献描述。本文提到了两种方法



方法 1: dm 处同时考虑 strain22 和 strain33 并取最大值, 计算方法如下,

$$dmt = \max(dmt_22, dmt_33)$$

$$dmc = \max(dmc_22, dmc_33)$$

那么可以理解为基体 22 方向或者基体 33 方向, 其中一个发生损伤, 那么基体会发生损伤。

方法 2: 如果 dm 分开考虑 22 方向和 33 方向, 即 dm_22, dm_33, 那么刚度退化时, C22 和 C33 是否也要分开考虑?

没有分开考虑如下:

$$C_{22} = (1 - d_f)(1 - d_m)C_{22}^0$$

$$C_{33} = (1 - d_f)(1 - d_m)C_{33}^0$$

分开考虑如下但是至今没有查到刚度退化分开考虑

$$C_{22} = (1 - d_f)(1 - d_{m,22})C_{22}^0$$

$$C_{33} = (1 - d_f)(1 - d_{m,33})C_{33}^0$$

如果采用此种方法, 那么基体 22 方向的损伤与 33 方向的损伤没有联系, 现实中, 如果基体损伤, 那么基体力学性能会下降, 那么 22 方向与 33 方向都会受到影响, 因此本文选取第一种方法。

3.4.8 l is the characteristic length of element.

问题: 运算 $\langle x \rangle = (x + |x|)/2$ 如何理解?

The symbol $\langle \rangle$ in the equations above represents the Macaulay bracket operator 麦考利函数, [abaqus analysis user's guide>materials>24 progress damage and failure>24.3.3 damage evolution and element removal for fiber-reinforced composites]

$$\langle x \rangle = \begin{cases} 0, & x \leq 0 \\ x, & x \geq 0 \end{cases}$$

characteristic length^[12] [7]

An accurate measure of the characteristic length is the ratio of the elemental volume V and fracture plane area A ,

$$l_{fib} = \frac{V}{A} \quad (10)$$

where A is calculated using an approach proposed in [21] and described in Section 2.1.2.

根据 abaqus 的帮助文档: charLength(nblock)表示单元的特征长度, 对于一



阶单元，四边形单元为面积的开方，六面体单元为体积的开三次方^[9]。

Abaqus user subroutines reference guide>1.2 abaqus/Explicit subroutines>1.2.22
VUMAT

charLength(nblock)

Characteristic element length, which is either the default value based on the geometric mean or the user-defined characteristic element length defined in user subroutine VUCHARLENGTH. **The default value is a typical length of a line across an element for a first-order element**; it is half of the same typical length for a second-order element. For beams, pipes, and trusses, the default value is a characteristic length along the element axis. For membranes and shells it is a characteristic length in the reference surface. For axisymmetric elements it is a characteristic length in the r-z plane only. For cohesive elements it is equal to the constitutive thickness.

3.5 损伤不可逆^[6]

To account for irreversibility, the damage variable as a function of analysis time, t , defined as

$$\begin{cases} d_{ii}(t+\Delta t) = \max\{0, \min\{1, d_{ii}(t+\Delta t)\}\} \\ d_{ii}(t+\Delta t) \geq d_{ii}(t) \end{cases}, i = 1, 2, 3. \quad (4)$$

对于初始损伤系数 $f_{ft}, f_{fc}, f_{mt}, f_{mc}, f_{ip}, f_{op}$
条件满足如下：

$$f = \max(f(t + \Delta t), f(t), 0)$$

对于刚度退化中的 damaged values d_{ij}

$$d = \max(d(t + \Delta t), d(t), 0) \\ d = [0, 1]$$



4 模型验证

4.1 材料参数

单位转换 $\text{J/m}^2 = \text{N/m} = \text{E-3 N/mm}$

Material properties of the carbon fibre/epoxy unidirectional laminate^[8]

Table 1

Material properties of the carbon fibre/epoxy unidirectional laminate [5,32].

Density (kg/m^3)	1600
Orthotropic properties	$E_1^0 = 153 \text{ GPa}$; $E_2^0 = E_3^0 = 10.3 \text{ GPa}$; $\nu_{12} = \nu_{13} = 0.3$; $\nu_{23} = 0.4$; $G_{12}^0 = G_{13}^0 = 6 \text{ GPa}$; $G_{23}^0 = 3.7 \text{ GPa}$
Strength (MPa)	$X^T = 2537$; $X^C = 1580$; $Y^T = 82$; $Y^C = 236$; $S_{12} = 90$; $S_{23} = 40$
In-plane fracture toughness (kJ/m^2)	$G_{1C}^T = 91.6$; $G_{1C}^C = 79.9$; $G_{2C}^T = 0.22$; $G_{2C}^C = 1.1$; $G_S = 0.7$

单位转换 $\text{KJ/m}^2 = \text{E3 J/m}^2 = \text{E3 N/m} = \text{N/mm}$

Material properties for T700GC/M21 composite laminates^[9]

Table 2

Material properties for T700GC/M21 composite laminates.

Composite lamina properties	Density	$P = 1600 \text{ kg/m}^3$
	Young's modulus	$E_{11} = 130 \text{ GPa}$, $E_{22} = E_{33} = 7.7 \text{ GPa}$, $G_{23} = 3.8 \text{ GPa}$, $G_{12} = G_{13} = 4.8 \text{ GPa}$
	Poisson's ratio	$\nu_{23} = 0.35$, $\nu_{12} = \nu_{13} = 0.33$
	Strength	$X_T = 2080 \text{ MPa}$, $X_C = 1250 \text{ MPa}$, $Y_T = 60 \text{ MPa}$ $Y_C = 140 \text{ MPa}$, $S_{12} = S_{13} = S_{23} = 110 \text{ MPa}$
	Fracture energy	$G_{ft} = 133 \text{ N/mm}$, $G_{fc} = 40 \text{ N/mm}$, $G_{mt} = 0.6 \text{ N/mm}$, $G_{mc} = 2.1 \text{ N/mm}$
Interface properties	Modulus	$E = 5 \text{ GPa}$
	Strength	$N = S = 30 \text{ MPa}$
	Fracture energy	$G_n^C = 0.6 \text{ N/mm}$, $G_s^C = 2.1 \text{ N/mm}$

4.2 Character length

$\text{charLength}(\text{nblock})$ 表示单元的特征长度，对于一阶单元，四边形单元为面积的开方，六面体单元为体积的开三次方^[9]，每个单元的体积为 3.125，开立方为 1.4620088



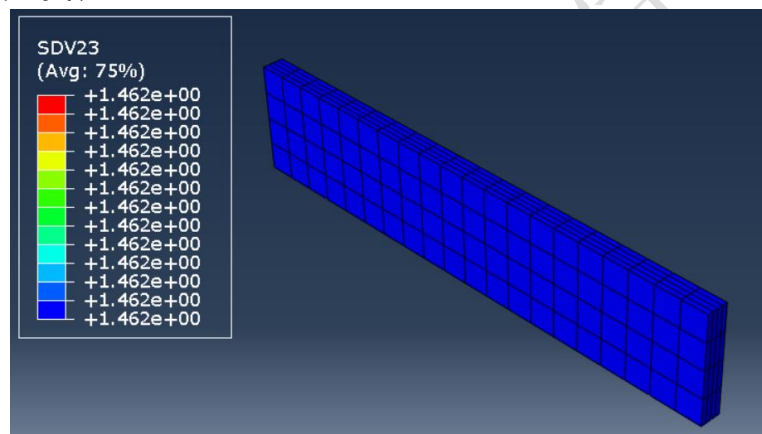
本文输出了每一步中的 `charlength(i)`，部分代码如下：

```
164 c ----- test for charLength(i) -----
165     stateNew(i,23) = charLength(nblock)
166     print *,i,' charLength is',charLength(nblock)
```

Log 文件中的部分结果如下：

```
1 charLength is 1.462009
2 charLength is 1.462009
3 charLength is 1.462009
***** 几个 frame 后
1 charLength is 1.454697
2 charLength is 1.454697
3 charLength is 1.454697
4 charLength is 1.454697
```

可以看出输出的第一步中，character length 是 1.42009，这与单元体积的开立方相同。随着分析步的计算，单元发生形变，体积发生变化，那么 character length 也发生变化



4.3 单元测试

通过建立 1 个单元或者很少的单元，测试子程序的顺利运行，根据输出判断是否准确。

模型中的输出应注意以下几点：

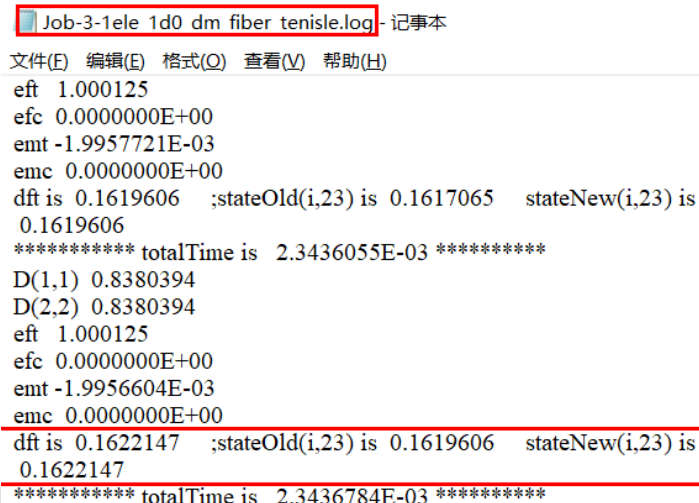
(1) Step field output 频率尽可能高，为了查看输出是否准确

(2) Step history output 频率尽可能高

VUAMT 中可以使用以下语句进行调试

(1) 使用 print 语句，具体格式如下，输出了 dft(纤维拉伸损伤系数)，结果可以在 log 文件中查看。注意 log 文件中是每一个分析步下所有点的计算输出，当节点较多时，log 文件会很大（几 GB），因此建议在测试时，用很少的单元测试。

```
print *, 'dft is ', dft, '; stateOld(i,23) is ', stateOld(i,23),  
& 'stateNew(i,23) is ', stateNew(i,23)
```



```
Job-3-1ele 1d0 dm fiber tensile.log - 记事本  
文件(F) 编辑(E) 格式(O) 查看(V) 帮助(H)  
eft 1.000125  
efc 0.0000000E+00  
emt -1.9957721E-03  
emc 0.0000000E+00  
dft is 0.1619606 ;stateOld(i,23) is 0.1617065 stateNew(i,23) is  
0.1619606  
***** totalTime is 2.3436055E-03 *****  
D(1,1) 0.8380394  
D(2,2) 0.8380394  
eft 1.000125  
efc 0.0000000E+00  
emt -1.9956604E-03  
emc 0.0000000E+00  
dft is 0.1622147 ;stateOld(i,23) is 0.1619606 stateNew(i,23) is  
0.1622147  
***** totalTime is 2.3436784E-03 *****
```

(2) 将重要变量设置为 statenew(i,j)，在 odb 中查看输出。

4.3.1 纤维拉伸

模型细节如下：

材料模型如下：



Material Behaviors

Density

Depvar

User Material

General Mechanical Thermal Electrical/Magn

Density

Distribution: Uniform

☐ Use temperature-dependent data

Number of field variables: 0

Data

	Mass Density
1	1.6E-009

Material Behaviors

Density

Depvar

User Material

General Mechanical Thermal Electrical/Magnetic Other

Depvar

Number of solution-dependent state variables: 48

Variable number controlling element deletion: 15



Material Behaviors

Density

Depvar

User Material

General

Mechanical

Thermal

Electrical/Magnetic

Other

User Material

User material type: Mechanical

☐ Use unsymmetric material stiffness matrix

Data

	Mechanical Constants
1	2537
2	1580
3	82
4	236
5	90
6	90
7	40
8	30
9	30
10	0.22
11	1.1
12	153000

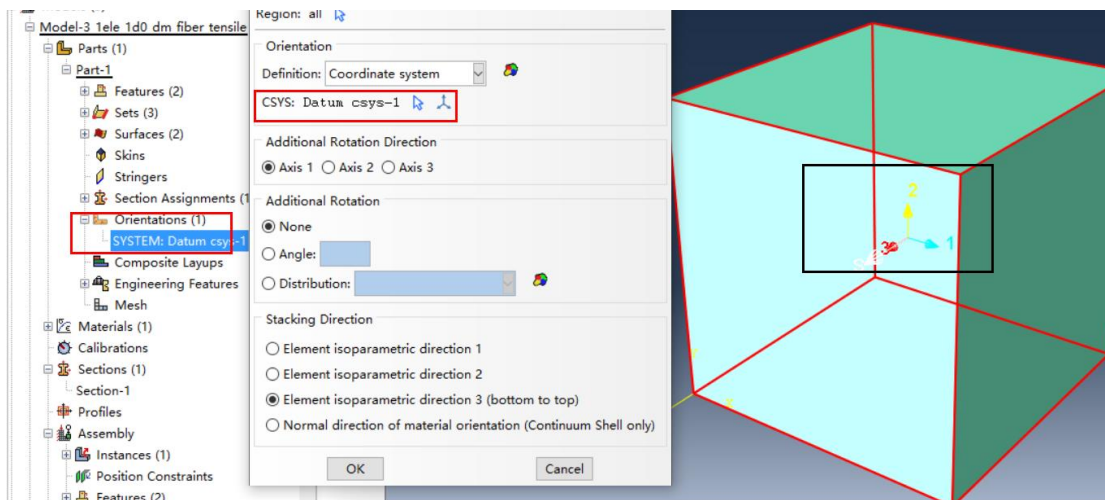
Section 如下:

Section Manager

Name	Type
Section-1	Solid, Homogeneous

材料方向如下:

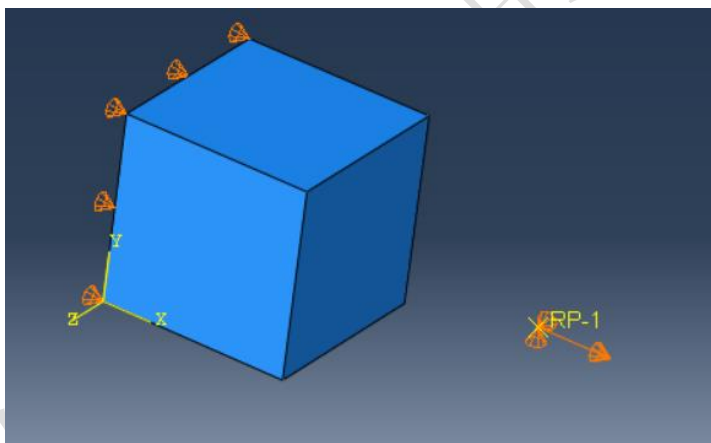




Step 信息如下:

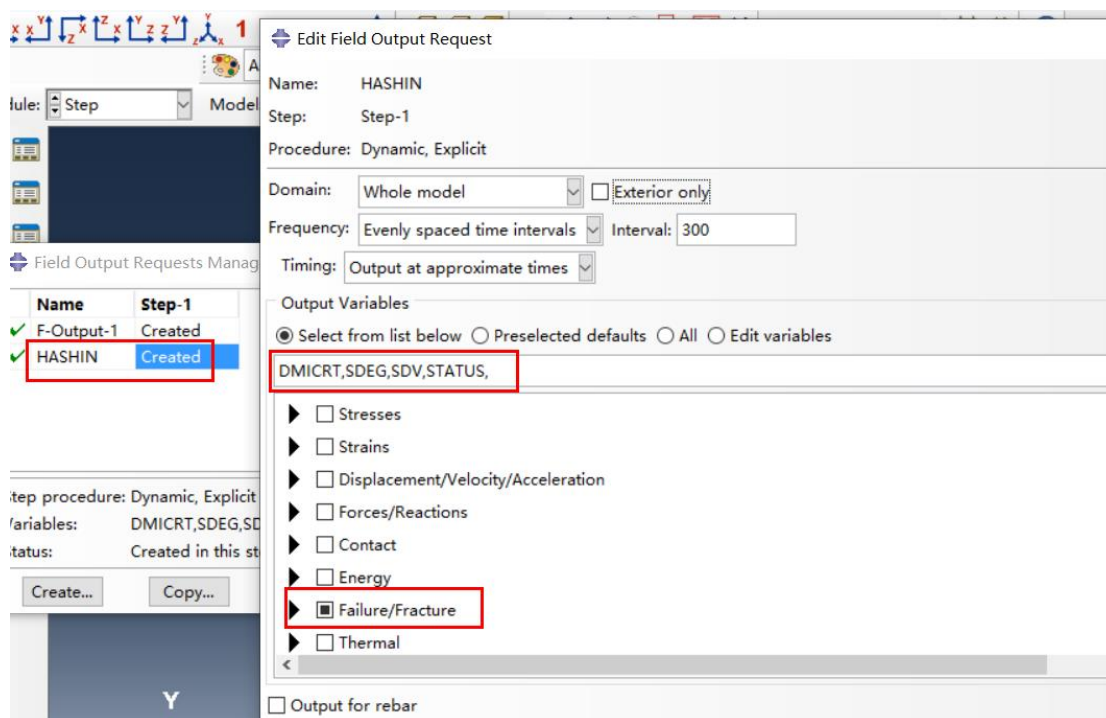
Step Manager			
Name	Procedure	Nlgeom	Time
Initial	(Initial)	N/A	N/A
Step-1	Dynamic, Explicit	ON	0.01

边界条件如下，左侧约束 1 方向位移，右侧在参考点施加 1 方向位移

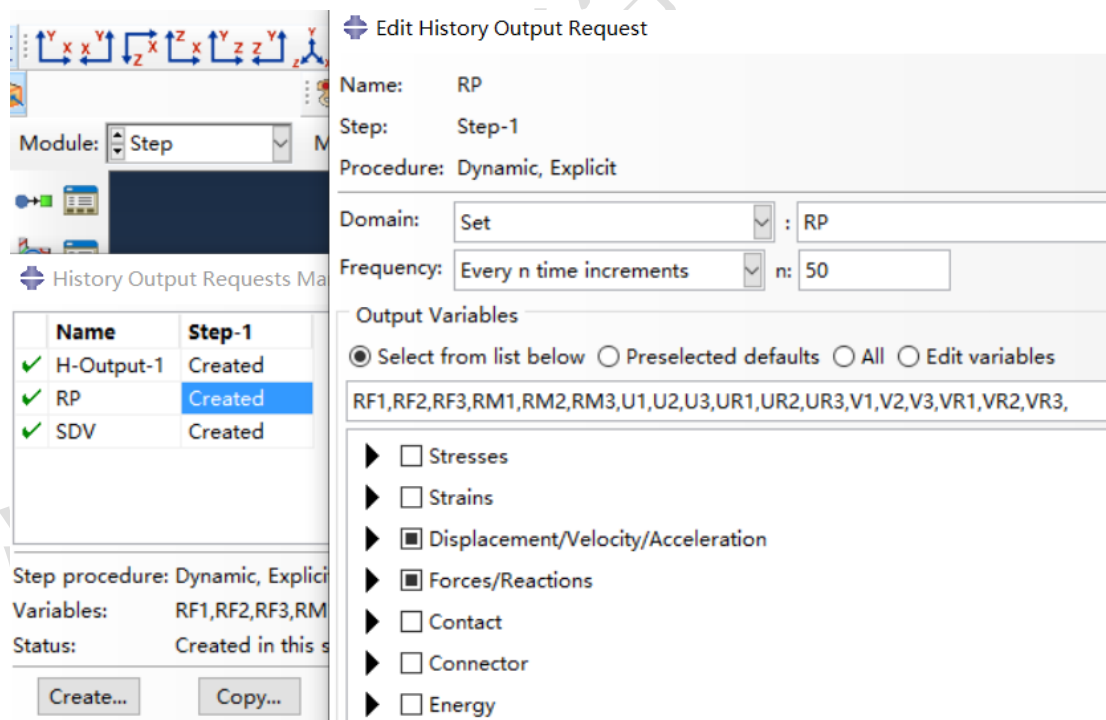


场输出如下，务必输出 STATUS。





历史输出如下：



计算结果如下

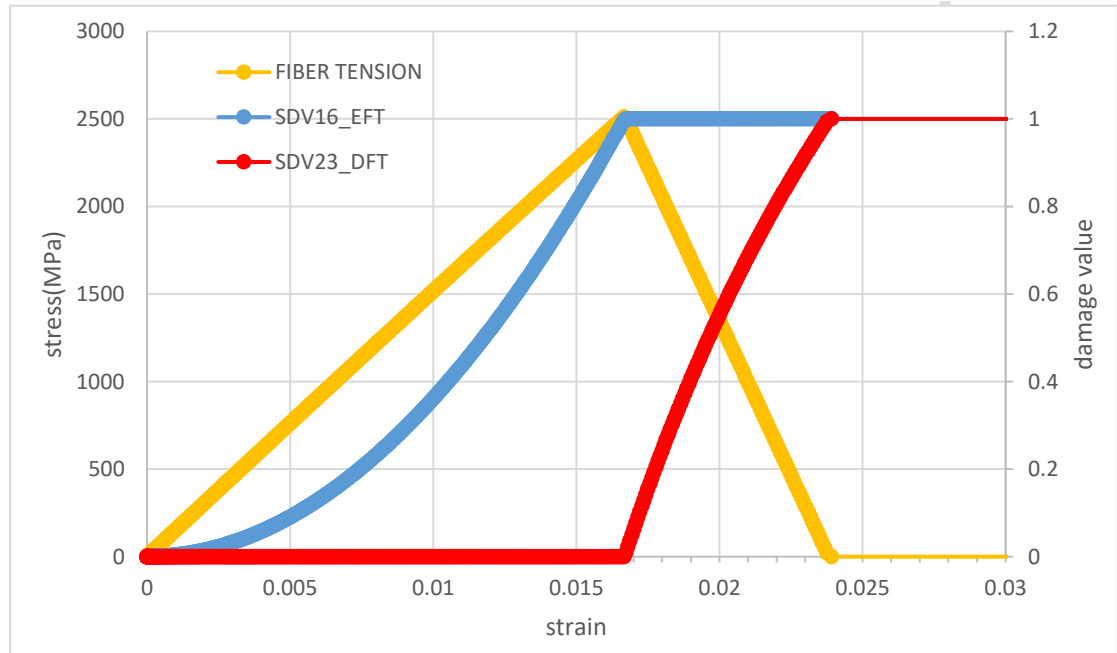
通过应力应变曲线可以看出：当应变为 0.0166661，应力为 2511.21MPa 时，发生初始损伤。峰值应力实验为 2537MPa，误差为-1.02%。

输出 SDV16 表示纤维拉伸的初始损伤系数 EFT，输出 SDV23 表示损伤演



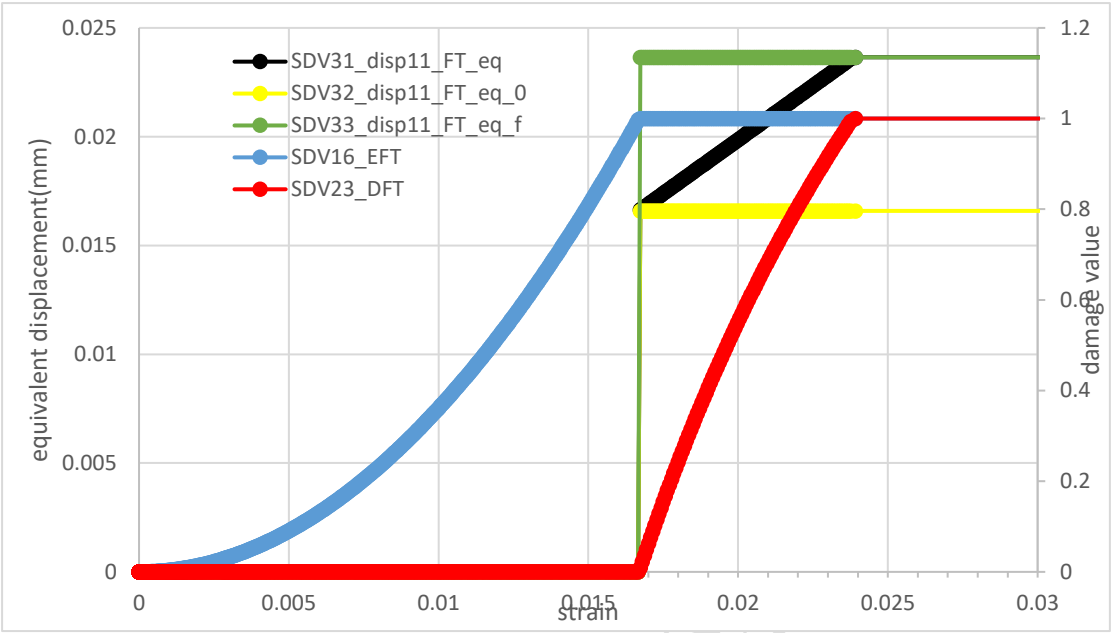
化过程中纤维拉伸的损伤系数 DFT。可以看出：随之位移的增大，EFT 从 0 逐渐增大；当应力达到 2511.21MPa 时，EFT=1，说明发生初始损伤，此时损伤系数 DFT 从 0 逐渐增大，刚度为线性退化。当达到能量释放率输入的 GFT 时，DFT=1，发生完全失效。

我们可以粗略计算图片中的能量释放率，单元尺寸为 1mm，因此特征长度为 1，那么图片中的能量释放率为 $2511.21 \times 0.0239153 \times 0.5 = 30.0281702565 \text{ N/mm}$ ，与输入的能量释放率 $GFT=30 \text{ N/mm}$ 相同，从而说明了模型的准确性。



如下图为损伤系数和计算的 11 方向等效位移：当初始损伤系数 EFT 达到 1 时，开始计算当前状态下的等效位移 δ_{eq} ，同时计算 δ_{eq}^0 (0.0165911) 和 δ_{eq}^f (0.02365)。随着位移的施加，等效位移逐渐增大，损伤系数 DFT 逐渐增大；当等效位移达到 δ_{eq}^f (0.02365) 时，DFT=1，说明发生了纤维拉伸失效。



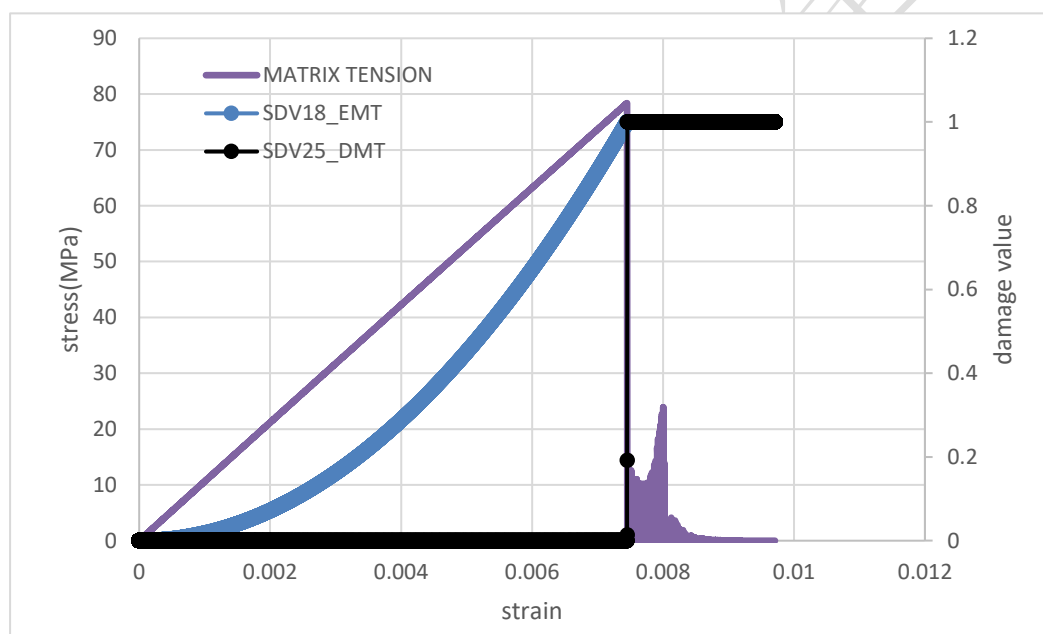


4.3.2 基体拉伸

通过更改局部坐标系，建立基体拉伸的单元测试案例。同时，单元尺寸与纤维拉伸不同，更改为 0.5mm。如果使用 1mm，那么计算的 δ_{eq}^0 大于 δ_{eq}^f 。因此减小单元尺寸。

应力应变曲线如下图：可以看出峰值应力为 78.3637MPa，实验值为 82MPa，误差为 4.43%。

对于损伤演化过程，随着应变的增大，应力逐渐增大，基体拉伸损伤系数 EMT 逐渐增大，当应力达到 78.3637MPa 时，EMT=1 说明发生了初始损伤失效，进入损伤演化阶段。随着应力的增大，损伤系数 DMT 逐渐增大，当达到 1 时，说明发生基体拉伸失效。



损伤演化过程较快，原因为基体拉伸的能量释放率很小（0.22N/mm），如下表，体现了损伤演化的过程。当 EMT=1 时，开始计算 DMT 和基体拉伸方向的等效位移。随着位移的施加，DMT 逐渐增大，当等效位移达到 δ_{eq}^f 时，DMT=1，说明发生基体拉伸失效。

单元为 0.5mm 的立方体，因此特征长度为 0.5mm，应变为 0.00745487，那么此时的位移为 0.003727435mm。78.3637*0.003727435*0.5=0.146，此值计算不准确，因为是整体的，应该看一个单元的应力应变状态。

查找此单元的应力和应变状态，最大 S22=78.5376，LE22=0.0113588

计算 22 方向 78.5376*(0.0113588*0.5)*0.5=0.223N/mm，与输入的 GMT(0.22) 相同，从而说明了模型的准确性。



U1	RF1	SDV18 _EMT	SDV25 _DMT	SDV37_di sp22_MT _eq	SDV38_di sp22_MT _eq_0	SDV39_di sp22_MT _eq_f	SDV43_di sp33_MT _eq	SDV44_di sp33_MT _eq_0	SDV45_di sp33_MT _eq_f
0.0074	78.3591	0.9998	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.0075	78.3637	1.0000	0.0005	0.0039	0.0039	0.0056	0.0003	0.0003	0.0979
0.0075	77.9722	1.0000	0.0141	0.0039	0.0039	0.0056	0.0003	0.0003	0.0979
0.0075	72.2053	1.0000	0.1918	0.0041	0.0039	0.0056	0.0003	0.0003	0.0979
0.0075	- 33.7983	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000



4.4 单轴拉伸

Material properties of the carbon fibre/epoxy unidirectional laminate^[8]

Table 1

Material properties of the carbon fibre/epoxy unidirectional laminate [5,32].

Density (kg/m ³)	1600
Orthotropic properties	$E_1^0 = 153 \text{ GPa}$; $E_2^0 = E_3^0 = 10.3 \text{ GPa}$; $\nu_{12} = \nu_{13} = 0.3$; $\nu_{23} = 0.4$; $G_{12}^0 = G_{13}^0 = 6 \text{ GPa}$; $G_{23}^0 = 3.7 \text{ GPa}$
Strength (MPa)	$X^T = 2537$; $X^C = 1580$; $Y^T = 82$; $Y^C = 236$; $S_{12} = 90$; $S_{23} = 40$
In-plane fracture toughness (kJ/m ²)	$G_{1C}^T = 91.6$; $G_{1C}^C = 79.9$; $G_{2C}^T = 0.22$; $G_{2C}^C = 1.1$; $G_S = 0.7$

单位转换 KJ/m²=E3 J/m²=E3 N/m=N/mm

模型尺寸如下：

Length 20

Width 4

Thickness 1

同时本文分别基于 EXPLICIT 和 STATIC, 使用 ABAQUS 自带的二维 hashin 进行计算, 输入的材料参数如下:

Edit Material

Name: carbon_2Dhashin_damage_explicit

Description:

Material Behaviors

Hashin Damage

Damage Evolution

Density

Elastic

General Mechanical Thermal Electrical/Magnetic Other

Hashin Damage

Alpha: 0

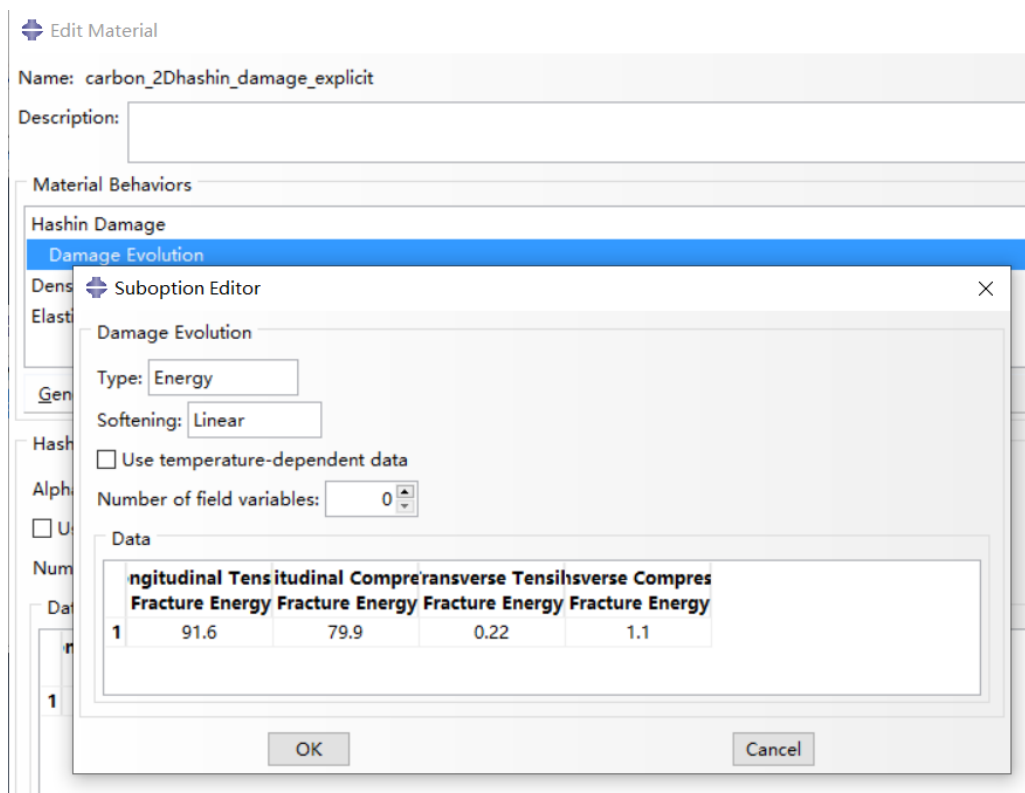
☐ Use temperature-dependent data

Number of field variables: 0

Data

	Longitudinal Tensile Strength	Longitudinal Compressive Strength	Transverse Tensile Strength	Transverse Compressive Strength	Longitudinal Shear Strength	Transverse Shear Strength
1	2537	1580	82	236	90	40

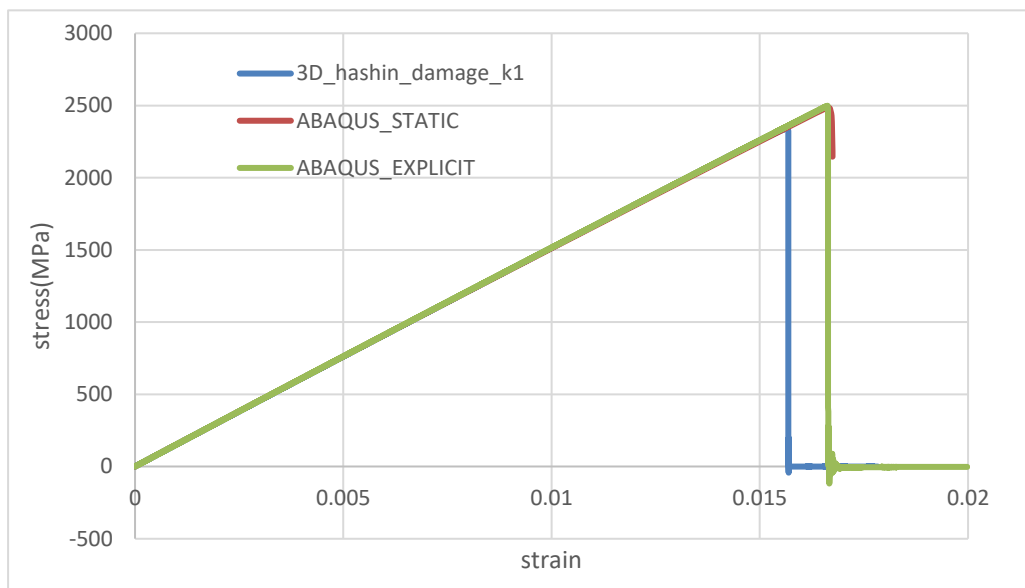




最终的计算结果如下，可以看出刚度误差为-0.35%，但是最大应力偏小，误差为-6.09%，失效应变也偏小，误差为-5.85%。原因为：过早的发生初始损伤，导致最大应力偏小。

	ABAQUS_STATIC	ABAQUS_EXPL	VUMAT	error
Stiffness	149443.1944	150283.095	149758.134	-0.35%
stress_max	2488.157959	2497.833984	2345.793457	-6.09%
failure strain	未计算完成	0.016662283	0.01568792	-5.85%





现在的纤维拉伸初始损伤判据为：

$$f_{ft} = \left(\frac{\sigma_{11}}{S_{11T}}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 + \left(\frac{\sigma_{13}}{S_{13}}\right)^2$$

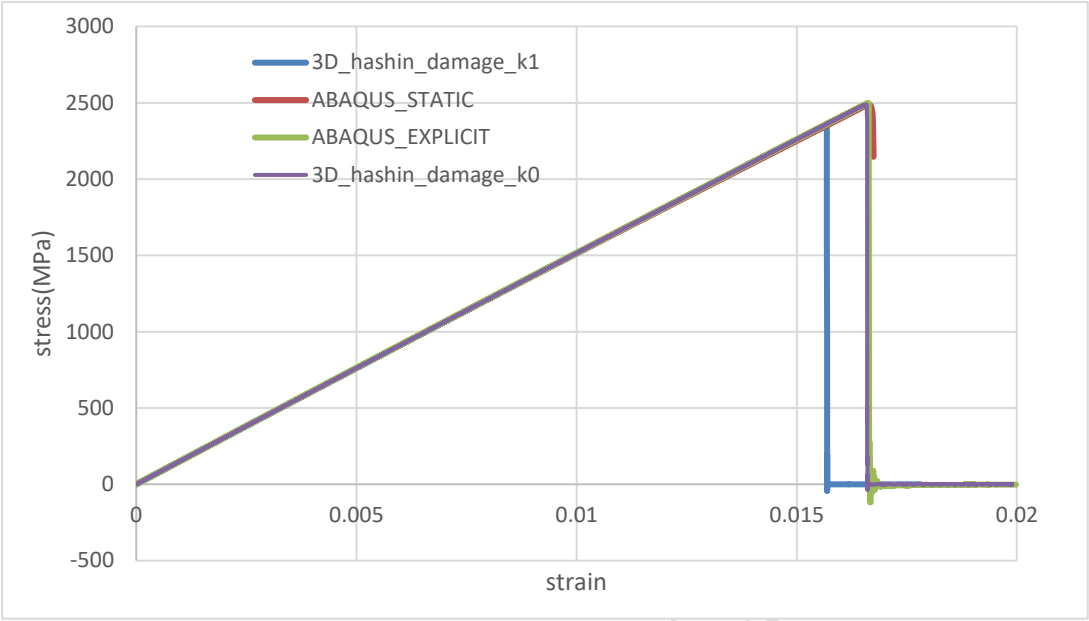
如果不考虑剪切的影响，如下，将程序中的参数 K 设置为 0.

$$f_{ft} = \left(\frac{\sigma_{11}}{S_{11T}}\right)^2$$

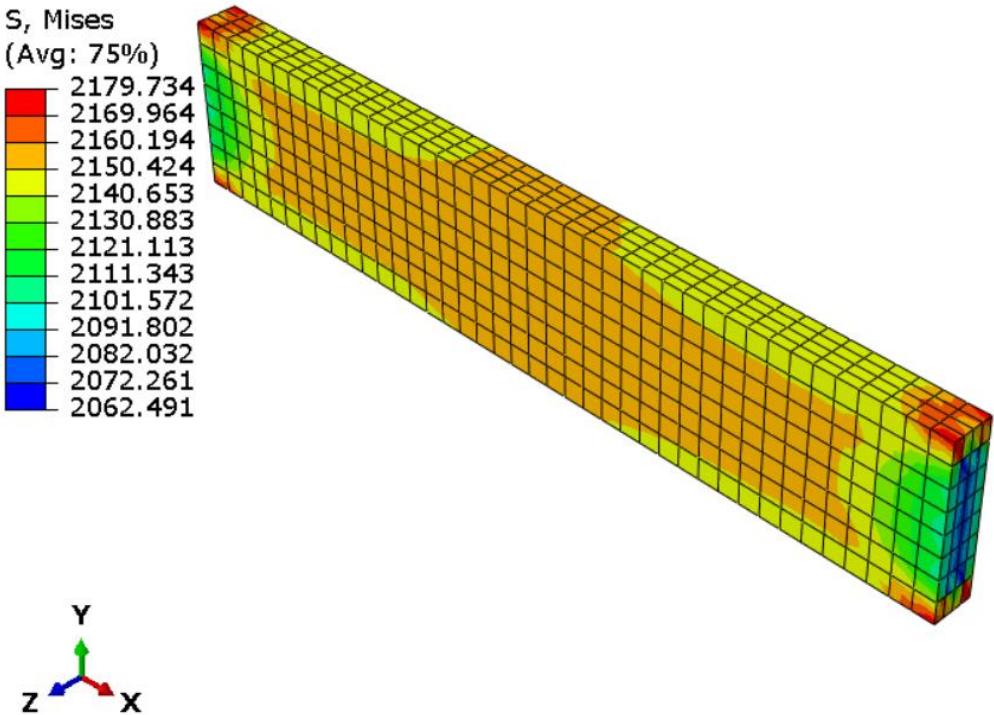
结果如下，可以看出刚度误差为-0.35%，最大应力误差为-0.38%，失效应变误差为-0.34%。

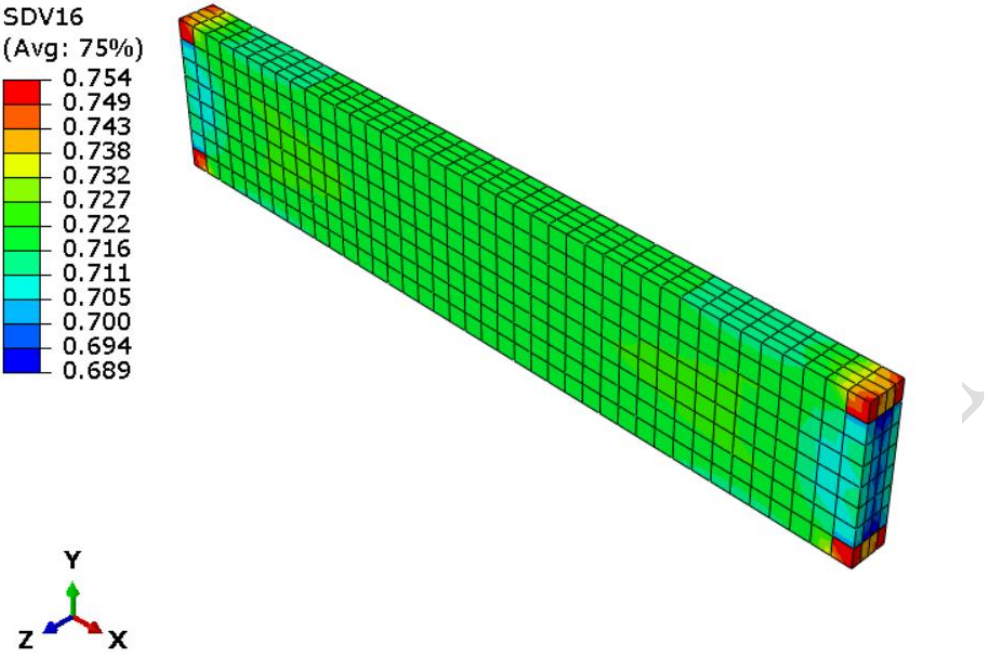
	ABAQUS_STATIC	ABAQUS_EXPL	VUMAT_k0	error
Stiffness	149443.1944	150283.095	150073.0422	-0.14%
stress_max	2488.157959	2497.833984	2488.34082	-0.38%
failure strain	未计算完成	0.016662283	0.016605058	-0.34%



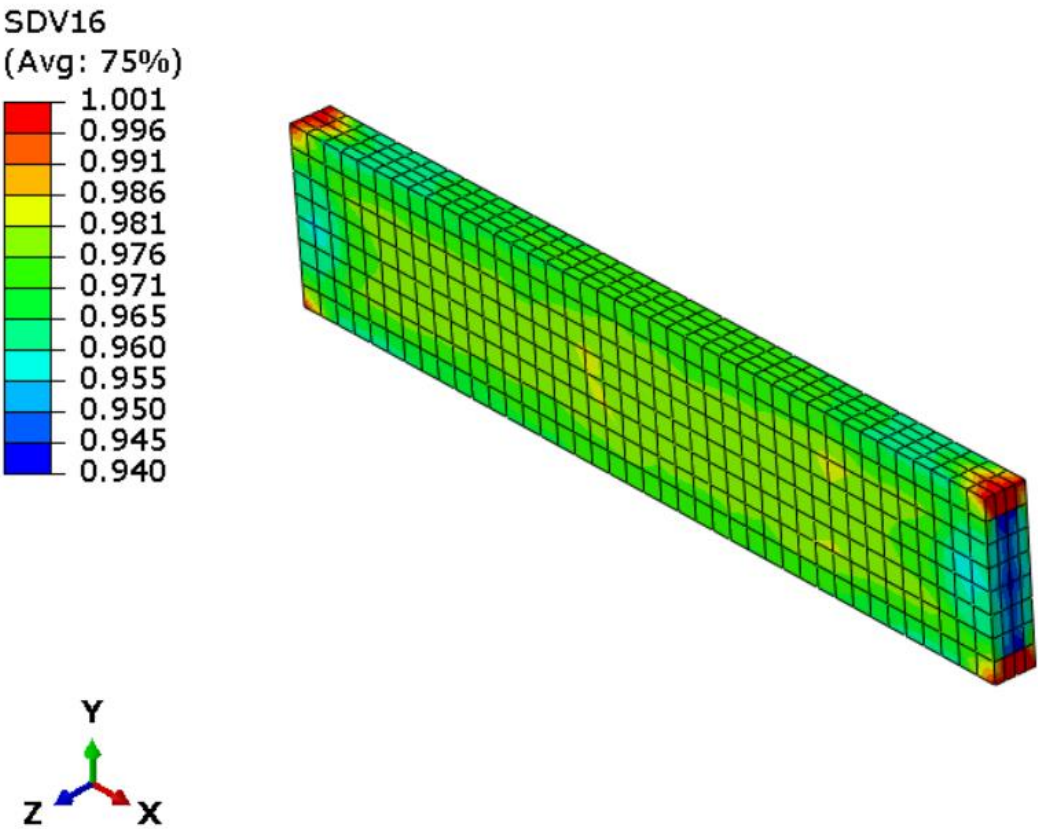


K=0 时，使用 VUMAT 的计算结果云图如下：

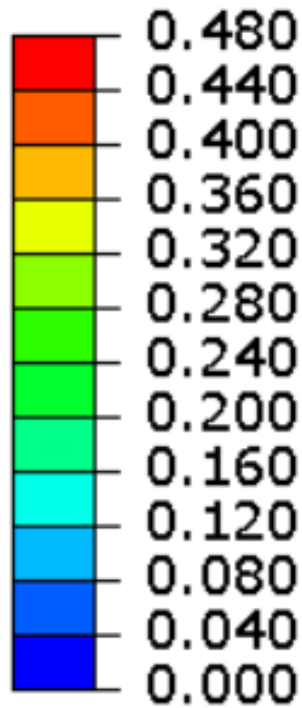




SDV16 EFT



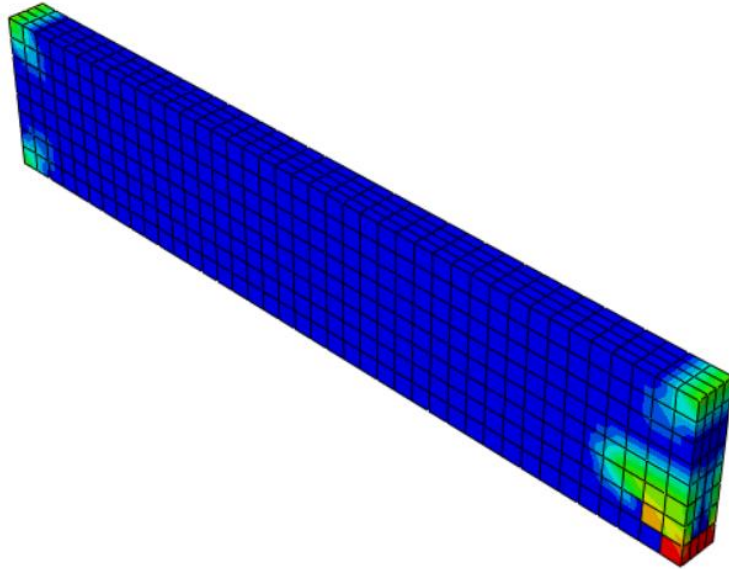
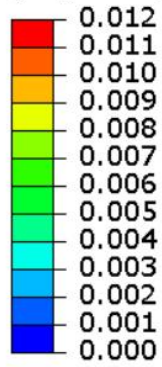
SDV23
(Avg: 75%)



SDV23 DFT



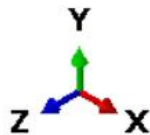
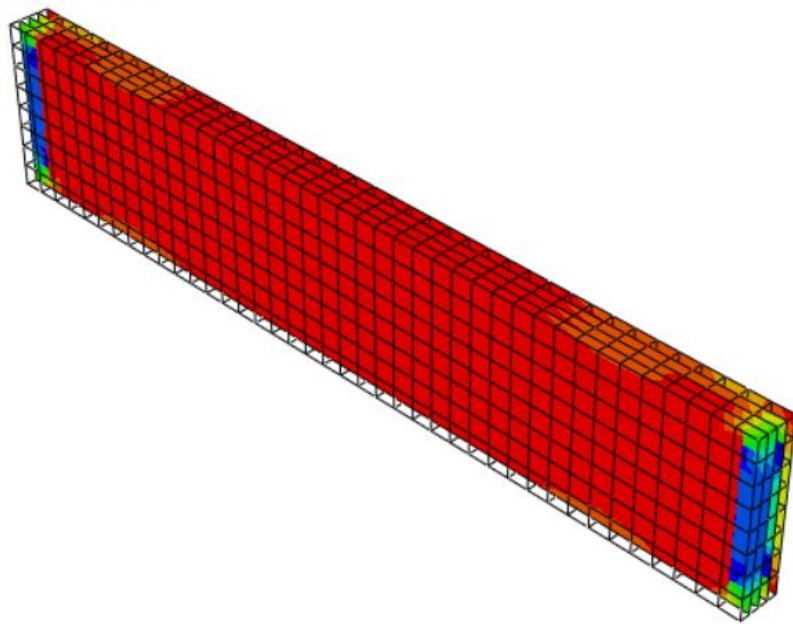
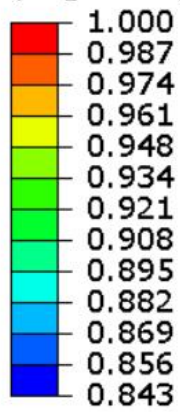
SDV31
(Avg: 75%)

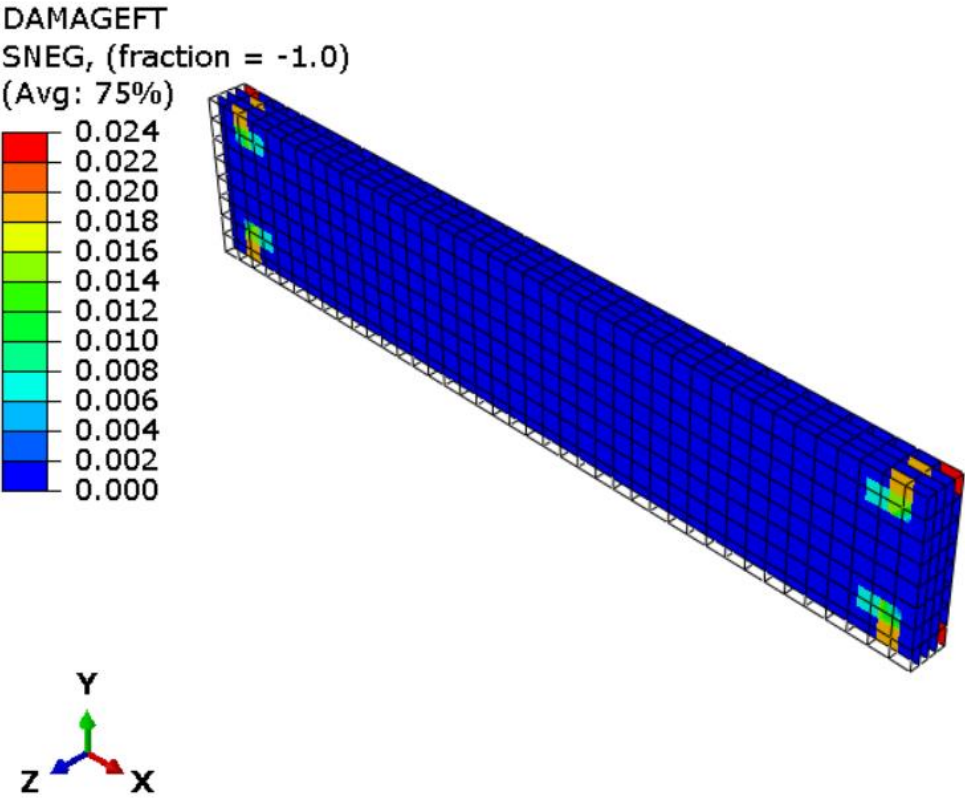


SDV 31 disp11_FT_eq

ABAQUS EXPL 计算云图如下:

HSNFTCRT
SNEG, (fraction = -1.0)
(Avg: 75%)





5 ABAQUS2022 与 2016 验证分析

改进前：在 abaqus2016 完成单精度测试

改进后：在 abaqus2022 完成单精度和双精度测试，解决单元畸变问题。

5.1 语法改进

Abaqus2016 与 2022 对 VUMAT 的单精度与双精度要求不同，如果使用本文 abaqus2016 的 VUAMT 会报错如下，原因是 maxval 函数中数组中的类型不同需要将函数中所有变量统一类型，因此本文需要统一所有变量的类型。

将 maxval 函数更改为 max 函数，max 函数对输入没有类型要求。

17_hashin_stress_progressive_damage_output.for(521): error #8209: If type specification is omitted, each element in an array-constructor must have the same type and kind type parameters. [0]

```
dft = maxval([dft,stateOld(i,23),0])
```

-----^

17_hashin_stress_progressive_damage_output.for(553): error #8209: If type specification is omitted, each element in an array-constructor must have the same type and kind type parameters. [0]

```
dfc = maxval([dfc,stateOld(i,24),0])
```

-----^

17_hashin_stress_progressive_damage_output.for(610): error #8209: If type specification is omitted, each element in an array-constructor must have the same type and kind type parameters. [0]

```
dmt = maxval([dmt_22,dmt_33,stateOld(i,25),0])
```

-----^

17_hashin_stress_progressive_damage_output.for(657): error #8209: If type specification is omitted, each element in an array-constructor must have the same type and kind type parameters. [0]

```
dmc = maxval([dmc_22,dmc_33,stateOld(i,26),0])
```

-----^

17_hashin_stress_progressive_damage_output.for(680): error #8209: If type specification is omitted, each element in an array-constructor must have the same type and kind type parameters. [0]

```
stateNew(i,j)=maxval([stateNew(i,j),stateOld(i,j),0])
```

另外，定义 SUBROUTINE 中的 2.d0 时，不要使用 two，因为 two 没有定义在 SUBROUTINE disp_eq_0_f 内，定义在外面。

```
SUBROUTINE disp_eq_0_f(disp_eq, stress_eq, eij, Gij, eq_0,eq_f)
```

```
real(kind=8), intent(in)::disp_eq,stress_eq,eij,Gij
```

```
real(kind=8), intent(out)::eq_0,eq_f
```



```

eq_0 = disp_eq / eij**0.5d0
eq_f = 2.d0*Gij / (stress_eq / eij**0.5d0)
end SUBROUTINE disp_eq_0_f

```

5.1 基体方向等效位移改进

对于基体拉伸与压缩破坏中的初始损伤等效位移和失效时的等效位移计算方法中，改进如下：

- (1) 如果发生基体拉伸损伤，执行第 2 步
- (2) 判断 2 方向的等效位移与 3 方向的等效位移大小
- (3) 选择等效位移大的计算初始损伤等效位移和失效时的等效位移，并用于计算损伤系数。

为何多了第二步中的判断哪个方向的等效位移大？

如果不判断，那么 2 方向和 3 方向均会计算初始损伤等效位移和失效等效位移。但是有可能只有一个方向发生了损伤，另外一个方向没有损伤，那么此时计算的另一个方向的初始损伤等效位移和失效位移错误。如下所示，采用 19 程序，进行 2 方向拉伸的单元测试，计算的 22 和 33 方向，可以看出 22 方向的失效位移为 **5.335569175068705E-003**，33 方向的失效位移为 **368311.905840240**，那么下一步计算的损伤系数中 $dmt_22 = 1.265332535024629E-004$, $dmt_33 = 0.975207652197101$, $dmt = 0.975207652197101$ 。因此 33 方向的损伤系数计算错误，最终计算的基体方向损伤系数错误。

```

***** find disp22_MT_eq_0 and disp22_MT_eq_f
stress_eq is 82.4655997836773
eij is 1.00000415724765
Gij is 0.219999998807907
two is 0.00000000E+00
one is 0.00000000E+00
zero is 0.00000000E+00
two*Gij is 0.000000000000000E+000
2.d0*Gij is 0.439999997615814
eij**0.5d0 is 1.00000207862166
stress_eq / eij**0.5d0 is 82.4654283692514
eq_f is 5.335569175068705E-003
stress_eq is 1.194642109669397E-006
eij is 1.00000415724765
Gij is 0.219999998807907
two is 0.00000000E+00
one is 0.00000000E+00
zero is 0.00000000E+00
two*Gij is 0.000000000000000E+000

```

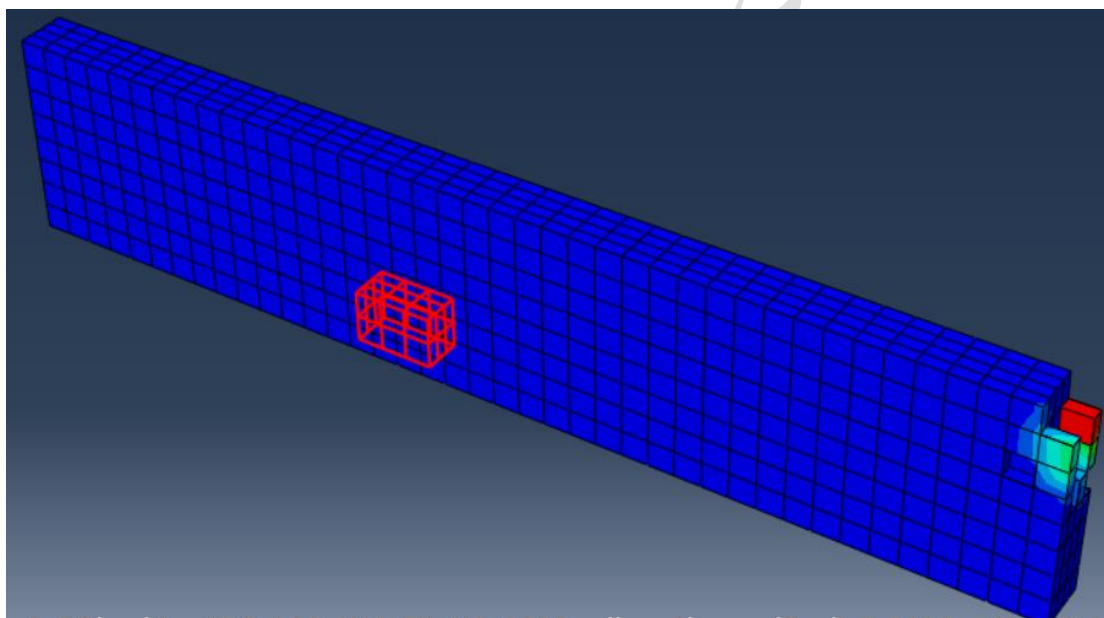


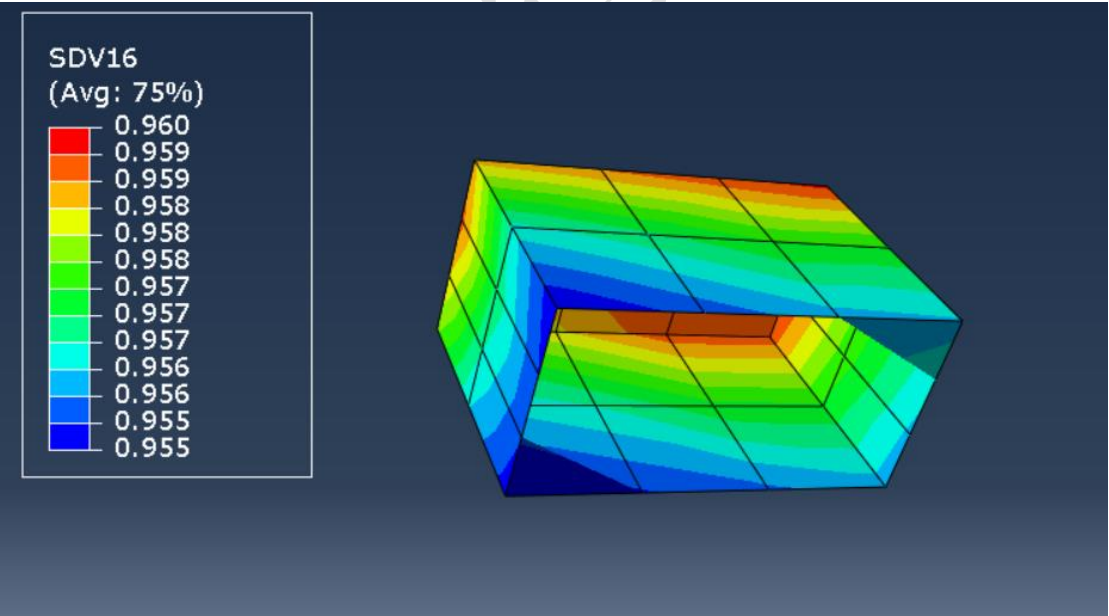
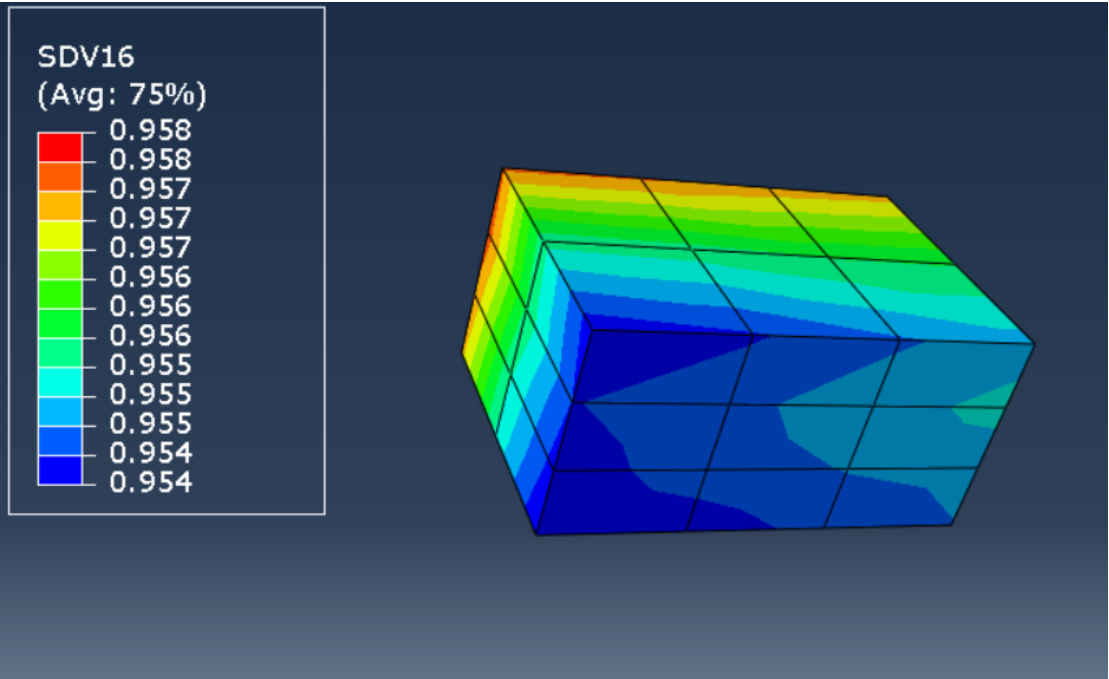
```

2.d0*Gij is 0.439999997615814
eij**0.5d0 is 1.00000207862166
stress_eq / eij**0.5d0 is 1.194639626465590E-006
eq_f is 368311.905840240
dmt_22 is 8.311315453535886E-006 dmt_33 is 2.078617342367146E-006
dmt is
8.311315453535886E-006 ;stateNew(i,25) is 8.3113155E-06
下一步中计算的损伤系数:
dmt_22 is 1.265332535024629E-004 dmt_33 is 0.975207652197101
dmt is 0.975207652197101 ;stateNew(i,25) is 0.9752076
改进后的 for 文件为:
20_hashin_stress_progressive_damage_2022.for
20_hashin_stress_progressive_damage_output_2022.for

```

5.2 单轴拉伸单元畸变问题





将 material 中的 statenew 输出数量改为和 for 文件相同，为 48。



5.3 Dij 改进

在损伤系数计算过程中，存在满足初始损伤后 ($e_{ij} \geq 1$)，由于受力的变化，部分单元受力急剧减小甚至为 0，那么此时计算的等效位移为 0，初始损伤等效位移为 0，失效等效位移为 0。那么程序会判定 `initailval .GE. finalval`，最终计算的 $d_{ij}=1$ ，发生失效。这与实际不同，因此需要增加等效位移是否为 0 的判断。

```
if ((now .EQ. 0.d0) .OR. (finalval .EQ. 0.d0)) then
  dij = 0.d0
end if
```

5.4 Eij 改进

问题描述：

部分单元删除后，剩余单元受力不稳定，发生剧烈变化。例如单元已经满足初始损伤并且发生了损伤，即 $e_{ij} \geq 1$ 。但是由于受力急剧变化，甚至拉压状态发生改变。例如 S11 由负数变为正数，原来计算的 $efc > 1$ ，变化后，应力 $S11 > 0$ ，将不会计算 efc ，而是计算 eft 。此时，程序中会赋值 $efc=0$ ，最然最终会将 `stateNew(i,17)` 进行损伤不可逆，`stateNew(i,17)` 的输出没有问题，但是程序中计算的 efc 仍然为 0。由于计算等效应变的前提条件时 $efc \geq 1$ ，那么后续将不会计算等效应变，导致当前增量步中的等效应变为 0。

解决方法：

计算完成 e_{ij} 后，进行损伤不可逆，并且不会被应力状态所限制。无论应力状态如何，均对 e_{ij} 进行损伤不可逆。代码如下：

```
if ( stress11 .GT. zero) then
  eft = (stress11 / Xt)**2.0+k*(stress12/S12)**2.0+
  &k*(stress13/S13)**2.0
c if the initail damage occurs , the damage is irreversible
C   if (stateNew(i,9) .GE. one) then
C     eft = max(eft,stateOld(i,16))
C   end if
else
  efc = (stress11 / Xc)**2.0
C   if (stateNew(i,9) .GE. one) then
C     efc = max(efc,stateOld(i,17))
C   end if
end if
c if the initail damage occurs , the damage is irreversible
if (stateNew(i,9) .GE. one) then
  eft = max(eft,stateOld(i,16))
  efc = max(efc,stateOld(i,17))
```



end if

御坂 sigma 的学习笔记



5.5 单元测试准确性分析

进行基体方向的拉伸，边长为 0.5 的立方体，共一个单元。

$RF1=20.5311N$

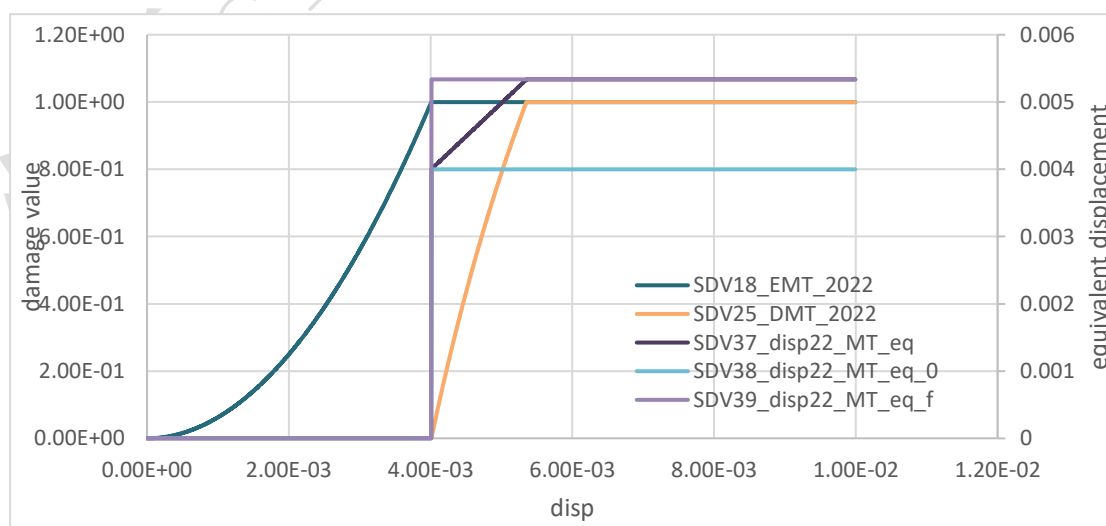
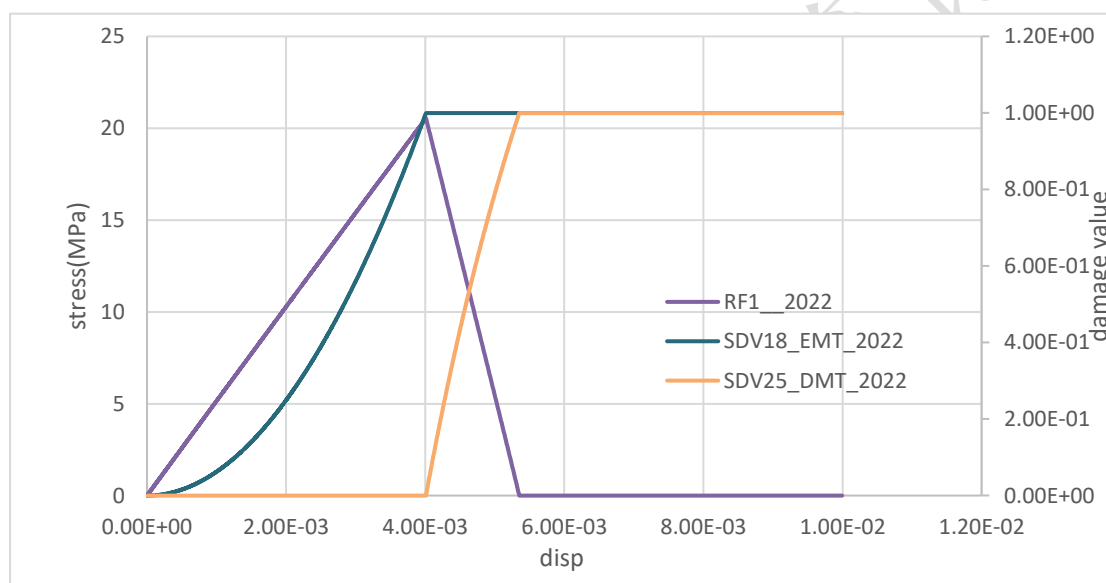
$U1=0.00534938mm$

$AREA=0.5*0.5=0.25$

$LC=0.5$

$STRESS=RF1/AREA=20.5311/0.25=82.1244$

能量释放率 $GMT=82.1244*0.00534938*0.5=0.2197$ 与输入的 0.22 相等



5.6 单轴拉伸准确性分析

	ABAQUS_STATIC	ABAQUS_EXPL	VUMAT_k0	error
Stiffness	149443.1944	150283.095	150431	0.1%
stress_max	2488.157959	2497.833984	2454.43	-1.74%
failure strain	未计算完成	0.016662283	0.01634	-1.91%



6 ref

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